Integral tail signatures and the Hilbert–Smith conjecture

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An expanded exposition

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1 Overview and main statements

Fix a prime p. The first statement concerns continuous actions of the additive group of p-adic integers, $\mathbb{Z}_p$, on finite-dimensional manifolds. All manifolds are Hausdorff and second countable, and may have boundary.

Theorem 1.1 (p-adic exclusion)**.** *Every continuous action of $\mathbb{Z}_p$ on a connected finite-dimensional manifold has nontrivial kernel.*

Corollary 1.2 (Hilbert–Smith)**.** *A second-countable locally compact Hausdorff group acting continuously and faithfully on a connected finite-dimensional manifold is a Lie group with its given topology.*

The terminology in these statements is developed in sections 2 and 3. The passage from p-adic exclusion to the corollary uses Newman's theorem and the structure theory of locally compact groups, recorded in section 16. The controlled construction itself is local: it uses a coordinate neighborhood in the original manifold and does not require an orientation or a triangulation of that manifold.

The two halves of the construction

The first half is an algebraic obstruction independent of any manifold action. Start with finite cyclic groups

$$G_i = C_{p^{a_i}}, \qquad P_i < G_i, \qquad [G_i : P_i] = p,$$

and a *single fixed* compact space T with a free action of $C_p = G_i/P_i$. For sequences of rational matrices labeled in T, require the nonzero entries to move labels by distances tending to zero as $i \to \infty$. These matrices form the controlled category $\mathcal{A}_{\mathbf{G}}(T)$. The obstruction is

$$\alpha \in L_4(\mathcal{A}_{\mathbf{G}}(T)) \quad \Longrightarrow \quad \sigma_{\mathrm{tail}}(\alpha) \text{ is eventually divisible by } p. \tag{1.1}$$

Its mechanism is local triviality of the C_p-covering $T \to T/C_p$, controlled Mayer–Vietoris, and a cyclotomic divisibility calculation. The L-groups are abelian groups over $\mathbb{Z}$, even though the chain coefficients are rational; consequently the conclusion really is divisibility of integers.

The second half constructs an incompatible class from a hypothetical faithful $\mathbb{Z}_p$-action. It produces the same fixed free C_p-space T and a class with

$$\alpha \in L_4(\mathcal{A}_{\mathbf{G}}(T)), \qquad \sigma_{\mathrm{tail}}(\alpha) = [(1, 1, 1, \ldots)]. \tag{1.2}$$

The contradiction is immediate once both statements have been established: no tail of 1's is divisible by p.

The objects constructed from an action

It is useful to distinguish the successive objects, since each solves a different problem:

$$\begin{aligned}\text{action on } M &\longmapsto \text{invariant sphere control } f : S^n \longrightarrow S^n,\\ &\longmapsto \text{free-action detector } \widetilde{\rho} : Z^+ \longrightarrow T,\\ &\longmapsto \text{finite chain models of } W = S^n \times \mathbb{CP}^2,\\ &\longmapsto \text{compressed chain complexes } C_i,\\ &\longmapsto \text{homotopy idempotents } E_i \text{ on } C_i \otimes \mathbb{Q}[G_i],\\ &\longmapsto \mathfrak{d} \in L_{n+4}(\mathcal{B}_{\mathbf{G},Z}(T \times S^n)),\\ &\longmapsto \alpha = \pm\Delta_T\mathfrak{d} \in L_4(\mathcal{A}_{\mathbf{G}}(T)).\end{aligned} \tag{1.3}$$

The auxiliary closed manifold W supplies a Poincaré chain complex; it does *not* endow M with a smooth or piecewise-linear structure. The sphere label f controls which chains are retained, while T records how group elements move those retained chains. The $\mathbb{CP}^2$ factor is chosen for a very specific reason: its middle-dimensional intersection form is $\langle 1\rangle$, of signature 1. Successive localization boundaries remove the sphere directions and leave exactly one copy of that form.

A finite quotient $G_i = H/K_i$ does *not* ordinarily act on M: the subgroup K_i need not fix any points. Choosing representatives in H instead gives chain maps whose multiplication laws hold only up to controlled homotopy. The free sheet factor $\mathbb{Q}[G_i]$ compensates their movement in T. Averaging over all sheets produces a duality-compatible homotopy idempotent; splitting it and applying the correct normalization recovers the original nonequivariant manifold class. The rest of the paper constructs these operations and checks their support, control, and duality properties.

2 Groups, actions, and p-adic averaging

2.1 Actions, kernels, and stabilizers

A *topological group* is a group whose multiplication and inversion are continuous. A *continuous left action* of a topological group G on a space M is a jointly continuous map

$$G \times M \longrightarrow M, \qquad (g, x) \longmapsto gx,$$

satisfying $1x = x$ and $g(hx) = (gh)x$. Each map $x \mapsto gx$ is a homeomorphism: its inverse is $x \mapsto g^{-1}x$.

For $x \in M$, the *orbit* and *stabilizer* are

$$Gx = \{gx : g \in G\}, \qquad G_x = \{g \in G : gx = x\}.$$

The *kernel* of the action is

$$\ker(G \curvearrowright M) = \{g : gx = x \text{ for all } x \in M\} = \bigcap_{x \in M} G_x.$$

An action is *faithful*, or equivalently *effective*, when its kernel is $\{1\}$. It is *free* when $G_x = \{1\}$ for *every* x. Thus a nonidentity element in a faithful action must move at least one point; in a free action it must move every point. A *free point* means a point with trivial stabilizer.

Example 2.1. The action of $C_2 = \{1, -1\}$ on $\mathbb{R}$ by multiplication is faithful but not free, because 0 is fixed. Every nonzero point is free. One must not replace faithfulness by freeness without an argument.

For a subgroup $H \leq G$, write

$$\mathrm{Fix}(H) = \{x : hx = x \text{ for every } h \in H\}.$$

If M is Hausdorff, G_x is closed in G, since it is the inverse image of the closed singleton $\{x\}$ under $g \mapsto gx$. Likewise $\mathrm{Fix}(H)$ is closed in M, as an intersection of closed equalizer sets. The action kernel is a closed normal subgroup. Normality follows from

$$(ghg^{-1})x = g(h(g^{-1}x)) = x$$

when h belongs to the kernel.

A map $F : X \to Y$ between G-spaces is *equivariant* if $F(gx) = gF(x)$. It is *invariant* if $F(gx) = F(x)$; this is equivariance for the trivial action on Y. An invariant map is constant on orbits and factors uniquely through the *orbit space* X/G, equipped with the quotient topology. That topology makes a function on X/G continuous precisely when its composite with $X \to X/G$ is continuous.

2.2 The p-adic integers

Fix a prime p. The *p-adic integers* form the inverse limit

$$\mathbb{Z}_p = \varprojlim_{k \geq 1} \mathbb{Z}/p^k\mathbb{Z}. \tag{2.1}$$

Concretely, an element is a compatible sequence $(x_k)_{k \geq 1}$, with $x_k \in \mathbb{Z}/p^k\mathbb{Z}$ and $x_{k+1} \equiv x_k \pmod{p^k}$. Addition and multiplication are coordinatewise. The paper uses the *additive* topological group underlying this ring.

Equivalently, an element has a unique digit expansion

$$x = a_0 + a_1 p + a_2 p^2 + \cdots, \qquad a_j \in \{0, \ldots, p-1\}.$$

This is convergence in the p-adic metric, not a real series. Put

$$v_p(x) = \min\{j : a_j \neq 0\}, \quad v_p(0) = \infty, \qquad d_p(x, y) = p^{-v_p(x-y)}.$$

The convention is $p^{-\infty} = 0$. Two elements are close when many of their initial digits agree. The metric satisfies the stronger triangle inequality

$$d_p(x, z) \leq \max\{d_p(x, y), d_p(y, z)\},$$

because divisibility by p^k is preserved under addition.

For $j \geq 0$, let

$$H_j = p^j\mathbb{Z}_p. \tag{2.2}$$

The cosets of H_j are open and closed, and the H_j form a neighborhood base at 0. Also

$$\mathbb{Z}_p/H_j \cong C_{p^j}, \qquad H_j/H_\ell \cong C_{p^{\ell-j}} \quad (\ell > j), \qquad \bigcap_j H_j = \{0\}.$$

Here C_m means a cyclic group of order m, for example $\mathbb{Z}/m\mathbb{Z}$. The notation $[G : P]$ denotes the number of left cosets of a subgroup P.

The space $\mathbb{Z}_p$ is compact: in (2.1) it is a closed subset of a product of finite discrete compact spaces. It is metrizable by d_p. It is *totally disconnected*: every connected subset is a singleton, since two different elements are separated by complementary clopen unions of cosets. A topological group expressed as an inverse limit of finite discrete groups is called *profinite*.

The ordinary integers embed densely in $\mathbb{Z}_p$. Indeed, the residue of x modulo p^k has an integer representative, giving an integer within p^{-k} of x. Multiplication by p^j is a topological group isomorphism $\mathbb{Z}_p \to H_j$.

Lemma 2.2 (Closed subgroups)**.** *Every nonzero closed subgroup of the additive group $\mathbb{Z}_p$ is H_j for some $j \geq 0$. Every proper closed subgroup of H_j is contained in H_{j+1}.*

Proof. Let S be a nonzero closed subgroup, and choose a nonzero $a \in S$ of minimum valuation j. Write $a = p^j u$, where u is a unit of $\mathbb{Z}_p$: its residue modulo p is nonzero, so its inverses modulo p^k form a compatible inverse in $\mathbb{Z}_p$. Density of $\mathbb{Z}$ implies

$$\overline{a\mathbb{Z}} = a\mathbb{Z}_p = p^j\mathbb{Z}_p = H_j.$$

Since S is closed and contains $a\mathbb{Z}$, it contains H_j. Minimality of j gives the reverse inclusion. The last assertion follows by applying this description inside H_j; the zero subgroup is also contained in H_{j+1}. □

The group $\mathbb{Z}_p$ has no nonzero element of finite order. To see this, write a nonzero integer $m = p^j u$ with u a unit; multiplication by u is invertible and multiplication by p^j is injective. Consequently $mx = 0$ implies $x = 0$. In particular, C_p is a *quotient* of $\mathbb{Z}_p$, not a subgroup consisting of elements of order p. This is one reason why a theorem about finite group actions does not automatically handle $\mathbb{Z}_p$-actions.

2.3 Haar measure: the averaging actually used

A *normalized Haar measure* on a compact group H is a Borel probability measure invariant under translation. For the groups here its construction is explicit. On $H = H_j$, every coset of H_{j+k} is assigned measure p^{-k}. These assignments are consistent: each coset splits into p cosets at the next level. They give the probability measure obtained by choosing the p-adic digits independently and uniformly.

Translation permutes the cosets at every level, so this measure is translation-invariant. Conversely, any translation-invariant probability measure must give all the p^k cosets equal measure p^{-k}; thus it is unique. Denote it by μ_H.

For a continuous vector-valued function $u : H_j \to \mathbb{R}^d$ one can calculate the integral by

$$\int_{H_j} u(h)\, d\mu_H(h) = \lim_{k\to\infty} \frac{1}{p^k} \sum_{r=0}^{p^k-1} u(p^j r). \tag{2.3}$$

Uniform continuity shows that using any representatives of the same cosets changes the sum by a quantity tending to zero. Translation invariance follows directly from this observation as well.

The four properties used later are normalization $\mu_H(H) = 1$, translation invariance, preservation of uniform bounds under averaging, and continuity with respect to a parameter. For the last property, if $u : H \times K \to \mathbb{R}^d$ is continuous with H, K compact, then

$$\left| \int_H u(h,x)\, d\mu_H - \int_H u(h,x')\, d\mu_H \right| \le \sup_{h \in H} |u(h,x) - u(h,x')| \longrightarrow 0.$$

The Haar measure is on the acting *group*, not on the manifold.

2.4 Uniform smallness and a free point

Lemma 2.3 (Small subgroups move compact sets uniformly little)**.** *For a continuous $\mathbb{Z}_p$-action on a metrizable space M and a compact subset $K \subset M$,*

$$\sup_{h \in H_j,\ x \in K} d(hx, x) \longrightarrow 0 \qquad (j \to \infty).$$

Proof. The map $(h, x) \mapsto d(hx, x)$ is continuous and vanishes on $\{0\} \times K$. For a prescribed $\varepsilon > 0$, continuity gives a product neighborhood of each $(0, x)$ on which the value is less than ε. Finitely many of the neighborhoods in K cover K. Intersecting their identity neighborhoods in $\mathbb{Z}_p$ and then taking H_j inside that intersection proves the claim. □

We will use the following manifold theorem, in exactly the form needed.

Foundational theorem 2.4 (Pointwise finite orbits; Roberts–Kaul–Yang)**.** *If a group acts by homeomorphisms on a connected manifold without boundary and every point orbit is finite, then the image of the group in the group of homeomorphisms is finite. In particular, a homeomorphism all of whose point orbits are finite has finite order. See [Rob75, Theorem 2].*

Lemma 2.5 (Faithfulness supplies a free point)**.** *A faithful continuous $\mathbb{Z}_p$-action on a connected manifold without boundary has a point with trivial stabilizer.*

Proof. Suppose every stabilizer were nonzero. By lemma 2.2, each stabilizer would contain some H_j, so the orbit of each point under the dense subgroup $\mathbb{Z} \subset \mathbb{Z}_p$ would be finite. Apply foundational theorem 2.4 to this integer action. There is an $m > 0$ whose action is the identity on the whole manifold. The closed action kernel therefore contains

$$\overline{m\mathbb{Z}} = p^{v_p(m)}\mathbb{Z}_p \neq \{0\},$$

contrary to faithfulness. □

3 The geometric and algebraic vocabulary

3.1 Manifolds, orientation, and the auxiliary space

A topological *n-manifold* is a Hausdorff, second-countable space locally homeomorphic to $\mathbb{R}^n$. "Hausdorff" means distinct points have disjoint neighborhoods; "second-countable" means there is a countable base of open sets. For manifolds with boundary, the local models also include the closed half-space $\mathbb{R}^{n-1} \times [0, \infty)$. A *chart* is one of these local homeomorphisms. A *closed*

manifold means a compact manifold without boundary, not merely a closed subset of another space. A *relatively compact* set has compact closure.

An *orientation* is a compatible choice of local fundamental generator. In homological language, at an interior point x of an n-manifold the local group $H_n(M, M \setminus \{x\}; \mathbb{Z})$ is $\mathbb{Z}$; the choices of generators must agree under restriction through small neighborhoods. A coordinate chart can be oriented even if M has no global orientation.

A map $c : S^n \to S^n$ has *degree* $d \in \mathbb{Z}$ when the induced map on top homology sends the fundamental generator to d times that generator. Degree is unchanged by a homotopy. A *homotopy* from c_0 to c_1 is a continuous $c : S^n \times [0,1] \to S^n$ with those endpoint maps. An orientation-preserving homeomorphism has degree 1. We use the sphere-map classification: for $n \geq 1$, two maps $S^n \to S^n$ are homotopic exactly when they have the same degree [Hat02, Corollary 4.25].

The sphere S^n can be viewed as $\mathbb{R}^n \cup \{\infty\}$, the *one-point compactification*: neighborhoods of ∞ are complements of compact subsets of $\mathbb{R}^n$. This compactifies the coordinate space used in the construction; it does not compactify the entire original manifold.

The complex projective plane is

$$\mathbb{CP}^2 = (\mathbb{C}^3 \setminus \{0\})/\mathbb{C}^*,$$

where nonzero vectors differing by a nonzero complex scalar are identified. It is a compact smooth manifold of real dimension 4. Its complex charts give a canonical orientation. Its middle homology is generated by a projective line $\mathbb{CP}^1$. Two distinct generic projective lines meet transversely in one point with positive sign, so the middle-dimensional intersection form is $\langle 1 \rangle$. In particular,

$$\sigma(\mathbb{CP}^2) = 1. \tag{3.1}$$

The auxiliary manifold is

$$W = S^n \times \mathbb{CP}^2, \qquad N = \dim W = n + 4,$$

with product orientation. All control maps used on W ignore its $\mathbb{CP}^2$ coordinate. The purpose of this factor is to leave a four-dimensional form of signature 1 after the n sphere directions have been removed by localization boundaries.

3.2 Triangulations and chains

A finite *simplicial complex* is a finite collection of simplices closed under taking faces, with intersections given by common faces. Its geometric realization is denoted $|K|$. A *triangulation* of a space is a homeomorphism from such a realization to the space. A *PL* map is piecewise linear after subdivisions. A *combinatorial triangulation* of an m-manifold is one in which simplex links are PL spheres or PL balls of the appropriate dimensions. In particular, its local stars have the familiar ball structure needed for dual cells.

The *mesh* of a triangulation in a metric space is the largest diameter of a closed simplex. Barycentric subdivision $\operatorname{sd} K$ replaces simplices by simplices whose vertices are barycenters of flags of faces. Iterated subdivision of a fixed finite triangulation has mesh tending to zero.

For a simplex σ, its *closed star* $\operatorname{st}_K \sigma$ is the subcomplex generated by all simplices containing σ. Its *link* consists of the faces disjoint from σ whose join with σ is a simplex. Closed stars are

contractible. If the mesh is η, any two points of $\mathrm{st}_K\,\sigma$ are at distance at most 2η, by comparison with a point of σ.

The rational simplicial chain group $C_r(K;\mathbb{Q})$ is the vector space with basis the oriented r-simplices. The boundary is

$$d[v_0,\dots,v_r]=\sum_{j=0}^{r}(-1)^j[v_0,\dots,\widehat{v}_j,\dots,v_r].$$

One has $d^2=0$. A *cycle* is an element of $\ker d$; a *boundary* is an element of $\operatorname{im} d$. Homology is $H_r(C)=\ker(d:C_r\to C_{r-1})/\operatorname{im}(d:C_{r+1}\to C_r)$. For a subcomplex $L\subset K$, the *relative complex* is $C_*(K,L)=C_*(K)/C_*(L)$.

The geometric *augmentation* is

$$\varepsilon:C_0(K;\mathbb{Q})\longrightarrow\mathbb{Q},\qquad \varepsilon\left(\sum_v a_v[v]\right)=\sum_v a_v.$$

If a zero-cell basis is replaced by signed basis elements, the augmentation includes those signs. A complex with an augmentation is *augmented acyclic* when its augmented sequence is exact. Equivalently, it has $H_0\cong\mathbb{Q}$ via augmentation and $H_r=0$ for $r>0$. Chains on a nonempty contractible complex have this property.

For a closed oriented triangulated m-manifold, the *fundamental chain* is the sum of oriented top simplices with coefficients ±1 determined by the orientation. Its boundary is zero. For a manifold A with boundary, orientations can be chosen so that

$$dz_A=jz_{\partial A},$$

where j is the inclusion of boundary chains and the boundary has the induced orientation.

3.3 Chain maps, homotopies, and cones

A *chain complex* is a sequence of vector spaces or modules C_r and maps $d_r:C_r\to C_{r-1}$ with $d_{r-1}d_r=0$. It is *bounded* when only finitely many C_r are nonzero. The notation $C\in[0,N]$ means $C_r=0$ outside that range. A *based* complex comes with a chosen basis in each degree.

A chain map $a:C\to D$ satisfies $d_Da=ad_C$. Two chain maps are *chain homotopic*, written $a\simeq b$, when

$$a-b=d_DH+Hd_C$$

for maps $H_r:C_r\to D_{r+1}$. A chain map is a *chain homotopy equivalence* if it has an inverse up to chain homotopy. A complex is *contractible* if its identity map is chain homotopic to zero.

For a chain map $j:C\to D$, the *mapping cone* is

$$\mathrm{Cone}(j)_r=D_r\oplus C_{r-1},\qquad d(x,c)=(d_Dx+jc,-d_Cc).\tag{3.2}$$

It records the relative homology of the map. A chain homotopy equivalence has contractible cone. For an inclusion of subcomplexes, the cone and the relative quotient are chain homotopy equivalent; an explicit controlled equivalence will be given in section 14.

Throughout the construction, the degree range is bounded independently of the index i. The dimensions of the chain groups need not be bounded: fine subdivisions and sheet copies produce arbitrarily large finite matrices.

3.4 Forms, projective modules, and signatures

A symmetric bilinear form $b : V \times V \to \mathbb{Q}$ is *nonsingular* if the map $V \to V^*$, $v \mapsto b(v, -)$, is an isomorphism. Its *signature* is

$$\sigma(V, b) = \dim V^+ - \dim V^-,$$

after extending scalars to $\mathbb{R}$ and splitting into positive and negative subspaces. This is an integer independent of the splitting. Orthogonal direct sum adds signatures; replacing b by $-b$ negates the signature. A positive scalar multiple has the same signature.

A *Lagrangian* is a subspace $L = L^{\perp}$; it is necessarily half-dimensional and the form vanishes on it. A form admitting a Lagrangian is *metabolic*. The *Witt group* consists of nonsingular forms modulo metabolic forms, with orthogonal sum as addition. A form and its negative sum to a metabolic form, so every class has an additive inverse. Signature descends to the Witt group because a metabolic real form has equally many positive and negative directions.

For a ring R, a *free module* is a module isomorphic to a direct sum of copies of R. A finitely generated module is *projective* if it is a direct summand of a finite free module. Equivalently, it is the image of an idempotent matrix e, where $e^2 = e$. Free modules and free group actions are different uses of the word "free." In our setting they are related: if B is a finite free G-set, then $\mathbb{Q}[B]$ is free over the group algebra $\mathbb{Q}[G]$, with one module generator per G-orbit in B.

The group algebra $\mathbb{Q}[G]$ consists of finite sums $\sum_g a_g g$ with convolution multiplication. Its involution is $\overline{\sum a_g g} = \sum a_g g^{-1}$. A *representation* of G is a linear action on a vector space. Its *character* at g is the trace of the linear map representing g. The complex representation ring $R_{\mathbb{C}}(G)$ is the group of formal differences of finite-dimensional complex representations, with tensor product as multiplication. Its elements are called *virtual representations*.

A ring is *semisimple* when its modules split as sums of simple modules; a nonzero module is simple if it has no nonzero proper submodule. The fact needed here is that $\mathbb{Q}[G]$ is semisimple for finite G. The splitting mechanism is elementary: average any linear projection onto a G-invariant subspace over G to obtain a G-equivariant projection. This is possible because $|G|$ is invertible in $\mathbb{Q}$.

4 Controlled categories and what equality means

4.1 The category of asymptotically small matrices

A *category* consists of objects, morphisms between them, associative composition, and identity morphisms. An *additive category* has abelian groups of morphisms, bilinear composition, a zero object, and finite direct sums. A *functor* preserves objects, morphisms, composition, and identities. It is *fully faithful* when it gives a bijection on every set of morphisms; this use of "faithful" concerns a functor, not a group action. An equivalence is a fully faithful functor for which every target object is isomorphic to an image object.

Fix finite cyclic groups $G_i = C_{p^{a_i}}$, index-p subgroups P_i, and specified identifications $G_i/P_i = C_p$. Let X be a compact metric C_p-space whose action is *isometric*, meaning $d_X(hx, hy) = d_X(x, y)$ for every group element h. We let G_i act on X through its quotient C_p.

Definition 4.1 (Asymptotic category)**.** An object of $\mathcal{A}_{\mathbf{G}}(X)$ is a sequence

$$M_i = \mathbb{Q}[B_i]$$

of finite based vector spaces, where B_i is a free G_i-set and the *label map* $\ell_i : B_i \to X$ is equivariant. A morphism $M \to M'$ is a sequence of G_i-equivariant rational matrices u_i, modulo eventual equality, satisfying

$$\operatorname{prop}(u_i) := \sup_{(u_i)_{b'b}\neq 0} d_X\big(\ell_i'(b'), \ell_i(b)\big) \longrightarrow 0. \tag{4.1}$$

The supremum over an empty set is 0. Eventual equality means equality of matrices for every sufficiently large i.

The labels are bookkeeping locations, not distinct points or a basis of functions on X. Several basis elements may have the same label. The action on the basis is free even when the action of G_i on labels has kernel P_i.

Propagation measures the largest distance moved by any nonzero matrix entry. It does not measure matrix norm or coefficient size. A coefficient $i!$ joining labels at distance $1/i$ is allowed. A coefficient $1/i$ joining labels at distance 1 is not asymptotically controlled. These distinctions are essential when we average over groups whose orders grow with i.

Matrix addition and composition obey

$$\operatorname{prop}(u_i + v_i) \le \max\{\operatorname{prop}(u_i), \operatorname{prop}(v_i)\}, \qquad \operatorname{prop}(v_i u_i) \le \operatorname{prop}(v_i) + \operatorname{prop}(u_i). \tag{4.2}$$

For the composition estimate, a nonzero entry in a product has at least one contributing intermediate basis element; apply the triangle inequality along that path. Thus the definition gives an additive category.

An *involution* or *duality* on this category reverses arrows: it identifies the dual basis with the original labeled basis and replaces each matrix by its transpose. Transposing does not change propagation, and transposes of equivariant matrices are equivariant because the group acts by permutation matrices. A functor is *duality-preserving* when it commutes with these duals, with the indicated natural identifications.

Forgetting equivariance gives scalar matrices. Pushing labels through a fixed continuous map $c : X \to Y$ preserves control: compactness gives uniform continuity, so

$$\operatorname{prop}(c_* u_i) \le \omega_c(\operatorname{prop}(u_i)) \longrightarrow 0,$$

where ω_c is a modulus of continuity. We use maximum metrics on products, so small propagation in a product means small propagation in each coordinate. The notation $\mathcal{A}_{\mathbf{1}}(X)$ means that all groups are trivial.

4.2 Supports and exterior quotients

For a closed invariant subset $S \subset X$, let $\mathcal{A}_S \subset \mathcal{A}_{\mathbf{G}}(X)$ be the full subcategory of objects satisfying

$$\sup_{b\in B_i} d_X(\ell_i(b), S) \longrightarrow 0. \tag{4.3}$$

Such objects are *supported asymptotically near* S. This does not require their labels to lie exactly in S. For $S = \varnothing$, take the zero subcategory.

Let $I_S(M, M')$ consist of morphisms factoring as $M \to N \to M'$ with $N \in \mathcal{A}_S$. Finite sums of such morphisms still factor through a direct sum of such objects. Thus I_S is an ideal: it is closed under addition and composition with arbitrary morphisms.

The *additive quotient* $\mathcal{A}/\mathcal{A}_S$ has the same objects as $\mathcal{A}$, with morphism groups

$$\mathrm{Hom}_{\mathcal{A}/\mathcal{A}_S}(M, M') = \mathrm{Hom}_{\mathcal{A}}(M, M')/I_S(M, M').$$

In particular, an S-supported object becomes zero because its identity factors through itself. We write

$$\mathcal{B}_{\mathbf{G},Z}(X) = \mathcal{A}_{\mathbf{G}}(X)/\mathcal{A}_Z. \tag{4.4}$$

The set Z will be an *exterior*: a region we intend to discard. This quotient retains a germ of the construction away from that region. It is not the category obtained merely by deleting labels in Z.

Lemma 4.2 (Endpoint test for the exterior ideal)**.** *A controlled morphism belongs to I_S exactly when both endpoints of every nonzero matrix entry approach S uniformly as $i \to \infty$.*

Proof. If $u = vw$ through an object with labels at distance at most $a_i \to 0$ from S, then every nonzero entry of u_i has a contributing intermediate label. Its source and target lie within $a_i + \mathrm{prop}(w_i)$ and $a_i + \mathrm{prop}(v_i)$ of S.

Conversely, keep just the source basis elements whose columns in u_i are nonzero. Equivariance makes this a union of free G_i-orbits. The resulting based submodule is S-supported, the projection to it has propagation zero, and the restriction of u supplies the second factor. □

Consequently there are two different exact notions of equality. In $\mathcal{A}$, matrices are equal after discarding finitely many indices. In $\mathcal{B}$, their difference may persist at every index, but must factor through an exterior-supported object using controlled factors. Neither notion is equality up to a small coefficient or small operator norm.

4.3 Support decompositions and excision

The relevant form of a *Karoubi filtration* is a full additive subcategory $\mathcal{S} \subset \mathcal{A}$ together with a directed family of decompositions of every object into an $\mathcal{S}$-part and a complement. The decompositions must be compatible with finite direct sums and sufficiently large to factor any finite collection of maps to or from $\mathcal{S}$ through the $\mathcal{S}$-parts. Here the decompositions are based and orthogonal, so duality preserves them.

For a null sequence $r_i \geq 0$, split the basis into labels at distance at most r_i from S and all remaining labels. The near summand belongs to $\mathcal{A}_S$. Given a map with propagation ε_i to an object within a_i of S, its nonzero source columns lie within $a_i + \varepsilon_i$ of S. The same argument handles maps from an S-supported object. Taking maxima handles a finite collection of maps and makes the family directed. This verifies the filtration property used by localization.

Proposition 4.3 (Support calculus)**.** *For a closed invariant cover $X = A \cup B$, with $Y = A \cap B$, one has*

$$\mathcal{A}_A \cap \mathcal{A}_B = \mathcal{A}_Y, \qquad \mathcal{A}_A/\mathcal{A}_Y \simeq \mathcal{A}_{\mathbf{G}}(X)/\mathcal{A}_B. \tag{4.5}$$

For nonempty closed invariant S, relabeling gives a duality-preserving equivalence $\mathcal{A}_{\mathbf{G}}(S) \simeq \mathcal{A}_S$. If $d(S, Z) > 0$, the functor $\mathcal{A}_S \to \mathcal{A}/\mathcal{A}_Z$ is fully faithful.

Proof. If points approach both A and B but not $A \cap B$, compactness supplies a convergent subsequence whose limit lies in both sets, a contradiction. This proves the intersection identity.

Split each basis according to labels in A and in $B \setminus A$. The second summand vanishes modulo $\mathcal{A}_B$, so the functor in (4.5) is essentially surjective. To check its kernel on morphisms, take a factorization between A-supported objects through a B-supported object. Retain only intermediate basis elements reached by the first map. They approach A by propagation and approach B by their support. Thus the factorization is through $\mathcal{A}_Y$. This proves full faithfulness.

For relabeling, choose one representative in each free basis orbit. Replace its label by a nearest point of S, and extend that choice equivariantly over the orbit. Freeness of the basis makes this unambiguous. The identity matrix between the old and new labels has propagation tending to zero, as does its inverse. Relabeling changes a matrix's propagation by at most the two labeling errors.

Finally, a morphism between S-supported objects that factors through Z has endpoints approaching both sets. Positive separation and the endpoint test force it to be eventually zero. □

4.4 Idempotent completion

An *idempotent* is a morphism $e : M \to M$ with $e^2 = e$. It *splits* if there are $D \xrightarrow{\iota} M \xrightarrow{r} D$ with $\iota r = e$ and $r\iota = 1_D$. An additive category need not contain the image of each idempotent as an object.

Its *Karoubi envelope*, or *idempotent completion*, $\operatorname{Kar}\mathcal{A}$ adds these images formally. An object is (M, e), and a morphism $(M, e) \to (N, f)$ is an original morphism $u : M \to N$ satisfying $u = fue$. The identity of (M, e) is e, and duality sends it to (M^*, e^*).

In controlled categories a formal summand (M, e) inherits the support of its ambient object M. The matrix e is still required to be a controlled morphism. We carry out based support decompositions before completion; there is no assumption that an arbitrary projective summand has a newly chosen geometric basis.

5 Poincaré complexes and the L-theory inputs

5.1 Dual chain complexes and signs

Write $C_r^* = \operatorname{Hom}_{\mathbb{Q}}(C_r, \mathbb{Q})$ for scalar complexes; in an additive category with duality, use that duality instead. For an integer N, the shifted dual complex is

$$(C^{N-*})_r = C^*_{N-r}, \qquad \delta_r = (-1)^r d^*_{N-r+1}. \tag{5.1}$$

These signs ensure $\delta^2 = 0$. A degree-zero map $\phi : C^{N-*} \to C$ has components $\phi_r : C^*_{N-r} \to C_r$. Its transpose in the shifted convention is

$$(T\phi)_r = (-1)^{r(N-r)} \phi^*_{N-r}. \tag{5.2}$$

The sign also identifies the double shifted dual with C. A map is *strictly symmetric* if $T\phi = \phi$, as an actual equality, rather than merely up to homotopy.

An N-dimensional *symmetric Poincaré complex* in the rational model used here is a bounded complex $C \in [0, N]$ with a strictly symmetric chain homotopy equivalence

$$\phi : C^{N-*} \longrightarrow C.$$

"Poincaré" refers to the resulting duality isomorphism $H^{N-r}(C) \cong H_r(C)$. The construction needs more than this homology isomorphism: ϕ, a chain inverse, and both inverse homotopies must be controlled. Uncontrolled linear algebra that merely proves the homology groups are isomorphic would not suffice.

For $N = 4k$, the middle-dimensional duality gives a symmetric form, and its signature is the ordinary signature of the Poincaré complex. In dimension 4, over a semisimple group algebra, algebraic surgery identifies the corresponding projective L-group with a Witt group of symmetric forms. This identification is one of the foundational inputs below.

5.2 Tensors, the diagonal, and the rational quadratic structure

For finite scalar complexes, the tensor product differential is

$$d(x_r \otimes y_s) = dx_r \otimes y_s + (-1)^r x_r \otimes dy_s.$$

The transposition is $T(x_r \otimes y_s) = (-1)^{rs} y_s \otimes x_r$. This is the *Koszul sign rule*: interchanging objects of degrees r, s contributes $(-1)^{rs}$.

The *right slant* of $x_r \otimes y_{N-r}$ is the map

$$\alpha \longmapsto x_r \alpha(y_{N-r}).$$

It identifies tensors of total degree N with maps $C^{N-*} \to C$. Under this identification, a tensor cycle gives a chain map, and tensor transposition gives (5.2).

For a general rational additive category, use the complex

$$\begin{aligned} \mathcal{M}(C)_k &= \bigoplus_r \mathrm{Hom}(C^*_{k-r}, C_r), \\ (\partial a)_r &= d_C a_{r+1} + (-1)^r a_r d^*_C, \\ (Ta)_r &= (-1)^{r(k-r)} a^*_{k-r}. \end{aligned} \tag{5.3}$$

One checks $T^2 = 1$ and $T\partial = \partial T$. Thus the invariant subcomplex consists of a with $Ta = a$, while the coinvariant complex identifies a and Ta.

The distinction between *symmetric* and *quadratic* structures normally involves additional chain-level data. Over $\mathbb{Q}$ it simplifies because 2 is invertible. Invariants and coinvariants for the involution are exact functors: averaging by $(1+T)/2$ is a projection onto invariants. The norm

$$1 + T : \mathcal{M}(C)_{C_2} \longrightarrow \mathcal{M}(C)^{C_2}$$

is an isomorphism with inverse sending an invariant tensor t to the coinvariant class of $t/2$. Consequently a strict symmetric structure ϕ has the quadratic representative

$$\psi_0 = \frac{1}{2}\phi, \qquad \psi_s = 0 \quad (s > 0). \tag{5.4}$$

The subscript s indexes the higher components in the general quadratic construction. They vanish in this rational representative. The *symmetrized* structure remains $(1+T)(\phi/2) = \phi$; the signature has not been divided by 2.

If invariant cycles differ by ∂H, the averaged primitive $(H + TH)/2$ has the same boundary. Halving that primitive identifies the quadratic structures as well. The same reasoning applies

to relative complexes and pairs because their structure maps commute with T [Ran80I, Section 3]. All these operations are finite sums, transposes, and scalar multiplications, so they preserve propagation estimates and exterior factorizations.

For an ordered triangulation, the *Alexander–Whitney diagonal* is the chain map

$$\Delta[v_0,\dots,v_m]=\sum_{r=0}^{m}[v_0,\dots,v_r]\otimes[v_r,\dots,v_m]. \tag{5.5}$$

Each term uses two faces of one simplex. For a fundamental chain z, put

$$t=\operatorname{Sym}\Delta z:=\tfrac{1}{2}(1+T)\Delta z.$$

If $dz=0$, then $dt=0$. Its slant is a strict symmetric cap map ϕ. The quadratic structure is $t/2$, or $\phi/2$. There are two appearances of $1/2$: the first symmetrizes the diagonal; the second passes from the symmetric structure to its quadratic lift.

5.3 Pairs, cobordism, and L-groups

A Poincaré *pair* is an algebraic version of a manifold with boundary. It consists of a chain map $j:P\to D$, boundary structure on P, and a relative structure on D. Its duality is an equivalence

$$\operatorname{Cone}(j)^{N-*}\longrightarrow D.$$

In the strict rational tensor model, the structure tensors satisfy

$$dt_D=j^{\otimes 2}t_P,\qquad dt_P=0.$$

The precise duality map, including signs, is given in section 14.

Two closed Poincaré complexes are *cobordant* if their disjoint union, with one structure negated, is the boundary of a Poincaré pair. Ordinary quadratic L-groups are the groups of such cobordism classes, with direct sum as addition. A Poincaré chain equivalence identifying structures up to the prescribed homotopy yields a cobordism by its algebraic mapping cylinder. This is why the chain equivalences constructed below identify L-classes, rather than only homology groups.

Different *decorations* on L-theory specify different finiteness and stabilization requirements. The superscript p in $L^p_k(R)$ means *projective*: finitely generated projective modules are allowed. Here that superscript is unrelated to the prime p. The decoration h uses free complexes and ordinary chain homotopy equivalences. The *ultimate lower* theory allows the successive lower stabilizations and is written

$$L_k(\mathcal{A}):=L_k^{\langle-\infty\rangle}(\mathcal{A})=\varinjlim_{j\to\infty}L_k^{\langle -j\rangle}(\mathcal{A}).$$

This fixes the convention of [Ran92, Definition 17.7]. The geometric complexes first define ordinary classes; the natural decoration maps place them in this ultimate group.

The L-groups can be assembled into a *spectrum* $\mathbf{L}^{\langle-\infty\rangle}(\mathcal{A})$. A spectrum packages homotopy groups in all integer degrees together with suspension maps; its kth stable homotopy group is $L_k(\mathcal{A})$. No explicit model of that spectrum is needed for the calculations below. Its role is to provide natural connecting maps with the correct degree shift. Suspension shifts homotopy groups by one: $\pi_k(\Sigma E)=\pi_{k-1}(E)$.

5.4 The foundational package actually invoked

Foundational theorem 5.1 (Localization and rational L-theory)**.** *The proof uses the following standard results from algebraic L-theory.*

(a) Functoriality and additivity. *A duality-preserving additive functor induces maps on L-groups; orthogonal direct sum gives addition. Tensoring with a positive permutation form is such a functor.*

(b) Karoubi localization. *For a duality-invariant Karoubi filtration $\mathcal{S} \subset \mathcal{A}$, there is a natural long exact sequence*

$$\cdots \to L_k(\mathcal{S}) \to L_k(\mathcal{A}) \to L_k(\mathcal{A}/\mathcal{S}) \xrightarrow{\partial} L_{k-1}(\mathcal{S}) \to \cdots . \tag{5.6}$$

"Exact" means that the image of each map is the kernel of the next. At the spectrum level this is a homotopy fiber sequence. Its relative-pair description sends a pair to the class of its boundary.

(c) Completion and decorations. *The natural map $L_k(\mathcal{A}) \to L_k(\operatorname{Kar}\mathcal{A})$ is an isomorphism. When all negative K-groups vanish, the comparison from projective to ultimate lower L-theory is an isomorphism. These comparisons are natural.*

(d) The semisimple coefficient calculation. *For a finite group G, the negative K-groups of $\mathbb{Q}[G]$ vanish, and*

$$L^p_4(\mathbb{Q}[G]) \cong \{\textit{nonsingular } G\textit{-invariant symmetric rational forms}\}/\textit{metabolic forms}.$$

The forgetful functor corresponds to the unnormalized coefficient-of-identity trace described in section 6.

(e) Finite support and filtered unions. *The relevant K- and L-constructions commute with filtered unions of additive categories in which each finite diagram lies in one stage. Thus a category with objects supported in finitely many sequence coordinates has L-groups given by the direct sum of the coordinate groups.*

Localization in (b) is [CP95, Theorem 4.2], together with the pair description in its proof of Theorem 4.1. The completion and comparison conventions are in [Ran92, Section 17]; the form calculation is in [Ran89, Theorem 5.2]. Negative K-group vanishing for semisimple rings is part of the regular-ring calculation in [Wei13, III, Section 4]. These are foundational inputs, not consequences of the elementary carrier arguments below.

For clarity about the term in (c), $K_0(R)$ is generated by classes of finitely generated projective modules with relations $[P \oplus Q] = [P] + [Q]$. The negative groups measure failures of stabilization. For example, Bass's definition is

$$K_{-1}(R) = \operatorname{coker}\left(K_0(R[t]) \oplus K_0(R[t^{-1}]) \longrightarrow K_0(R[t, t^{-1}])\right),$$

with higher negative groups obtained by repeating the construction. Polynomial matrices, idempotents, and their finite relations account for the finite-stage assertion needed in (e). We will not assume that L-theory commutes with an arbitrary infinite product.

A second input concerns homotopy idempotents. The *bounded homotopy category* $K^b(\mathcal{A})$ has bounded complexes as objects and chain maps modulo chain homotopy as morphisms.

Foundational theorem 5.2 (Splitting in the bounded homotopy category)**.** *For an idempotent-complete additive category $\mathcal{E}$, the category $K^b(\mathcal{E})$ is idempotent complete. In particular, a chain map E with $E^2 \simeq E$ splits in $K^b(\operatorname{Kar}\mathcal{A})$. This follows from the bounded derived-category splitting theorem of [BS01, Theorem 2.8], applied to the split exact structure.*

Here "split exact structure" means that only short exact sequences isomorphic to direct-sum sequences are used; for this structure the bounded derived category is the bounded homotopy category. No uncontrolled coordinatewise splitting will be substituted for this category-level statement.

6 Signatures, characters, and tails

6.1 The unnormalized trace

Let G be finite and (V, b) a nonsingular invariant rational form. Invariance means $b(gv, gw) = b(v, w)$. Its corresponding group-algebra-valued form is

$$\lambda(v, w) = \sum_{g \in G} b(v, gw)g. \tag{6.1}$$

The coefficient-of-identity map $\mathrm{tr}_1 : \mathbb{Q}[G] \to \mathbb{Q}$ sends $\sum a_g g$ to a_1, so

$$\mathrm{tr}_1 \, \lambda(v, w) = b(v, w).$$

There is no factor $1/|G|$. For example, $\lambda(hv, w) = h\lambda(v, w)$ and $\lambda(v, hw) = \lambda(v, w)h^{-1}$, as required by the involution on the group algebra. This is the convention compatible with forgetting equivariance and keeping transpose duality.

6.2 The signature representation

Choose a G-invariant positive-definite real inner product on $V_{\mathbb{R}}$. It exists by averaging an arbitrary positive inner product over the finite group. Write $b(v, w) = \langle Sv, w\rangle$ with S self-adjoint. Invariance implies that S commutes with G, so its positive and negative spectral subspaces V^+, V^- are G-invariant. Define

$$\mathrm{Sign}_G(V, b) = [V^+_{\mathbb{C}}] - [V^-_{\mathbb{C}}] \in R_{\mathbb{C}}(G). \tag{6.2}$$

This does not depend on the chosen inner product. One way to see this is to project one maximal positive subspace along a negative subspace onto another positive subspace. The projection is a G-equivariant isomorphism; its kernel would otherwise contain a vector that is both positive and negative for b.

If a form has a G-invariant Lagrangian L, average a projection to obtain a G-invariant complement K. The pairing $L \times K \to \mathbb{R}$ is nonsingular. For $u \in K$, define $c(u) \in L$ by $b(c(u), v) = b(u, v)$ for $v \in K$. Replacing u by $u - c(u)/2$ makes K Lagrangian. The resulting hyperbolic form has isomorphic positive and negative G-representations, so its signature representation is zero. Thus (6.2) is a Witt invariant.

Let $\chi_{\mathrm{Sign}_G(V,b)}(g)$ be its character. At the identity,

$$\chi_{\mathrm{Sign}_G(V,b)}(1) = \sigma(V, b) \in \mathbb{Z}. \tag{6.3}$$

Tensoring with a positive G-representation U multiplies characters:

$$\mathrm{Sign}_G(U \otimes V) = U_{\mathbb{C}} \otimes \mathrm{Sign}_G(V).$$

Positivity is important: the positive and negative spaces are exactly $U_{\mathbb{R}} \otimes V^+$ and $U_{\mathbb{R}} \otimes V^-$.

Lemma 6.1 (The cyclotomic divisibility step)**.** *Let $G = C_{p^a} = \langle g \rangle$. If $\chi_{\mathrm{Sign}_G(V,b)}(g) = 0$, then $p \mid \sigma(V, b)$.*

Proof. The complex irreducible representations of a cyclic group are one-dimensional. If ζ is a primitive p^ath root of unity, their characters at g are $1, \zeta, \ldots, \zeta^{p^a-1}$. Consequently

$$\mathrm{Sign}_G(V, b) = \sum_{j=0}^{p^a-1} m_j \chi_j, \qquad m_j \in \mathbb{Z}.$$

Let $F(t) = \sum_j m_j t^j \in \mathbb{Z}[t]$. The hypothesis says $F(\zeta) = 0$. The minimal polynomial of ζ is the cyclotomic polynomial

$$\Phi_{p^a}(t) = 1 + t^{p^{a-1}} + \cdots + t^{(p-1)p^{a-1}}.$$

It divides F in $\mathbb{Q}[t]$; since it is monic, polynomial division shows that the quotient lies in $\mathbb{Z}[t]$. Evaluating at 1 gives

$$\sigma(V, b) = F(1) \in \Phi_{p^a}(1)\mathbb{Z} = p\mathbb{Z}.$$

□

6.3 Why a sequence of integers is the invariant

An integer *tail* is a class in

$$\mathbb{Z}^{\mathbb{N}}/\mathbb{Z}^{(\mathbb{N})},$$

where $\mathbb{Z}^{\mathbb{N}}$ is the group of all integer sequences and $\mathbb{Z}^{(\mathbb{N})}$ is the subgroup of finitely supported sequences. Two sequences give the same tail exactly when they agree eventually. This is not a quotient by sequences converging to zero in a real norm.

A tail is *eventually divisible by p* if a representative (s_i) satisfies $s_i \in p\mathbb{Z}$ for all sufficiently large i. This property is independent of the representative. The constant tail $[(1, 1, \ldots)]$ is not eventually divisible by any prime.

Forgetting spatial labels gives a functor to the category

$$\mathcal{V}_{\mathbf{G}} = \mathcal{P}/\mathcal{S}, \qquad \mathcal{P} = \prod_i \mathrm{Free}(\mathbb{Q}[G_i]), \tag{6.4}$$

where $\mathcal{S}$ consists of objects with only finitely many nonzero coordinates. Morphisms in this quotient are arbitrary sequences of equivariant matrices modulo eventual equality; there is no spatial propagation condition.

Proposition 6.2 (Tail extraction)**.** *There is a natural map*

$$L_4(\mathcal{V}_{\mathbf{G}}) \longrightarrow \frac{\prod_i L_4^p(\mathbb{Q}[G_i])}{\bigoplus_i L_4^p(\mathbb{Q}[G_i])}. \tag{6.5}$$

It commutes with forgetting equivariance and with tensoring by positive permutation forms.

Proof. Splitting a sequence into a finite block of coordinates and its complement is a duality-invariant Karoubi filtration. The finite-support category is the filtered union of finite products of the coordinate categories. Each finite collection of polynomial matrices, idempotents, and relations involved in a negative K-group calculation lies in a finite block. Since the coordinate

rings are semisimple, the negative groups of $\mathcal{S}$ vanish. The foundational comparisons therefore give

$$L_j(\mathcal{S}) = \bigoplus_i L_j^p(\mathbb{Q}[G_i]), \qquad L_j(\mathrm{Free}(\mathbb{Q}[G_i])) = L_j^p(\mathbb{Q}[G_i]).$$

The map $L_3(\mathcal{S}) \to L_3(\mathcal{P})$ is injective: projecting back to each coordinate detects every entry of a finitely supported class. Exactness of localization now makes $L_4(\mathcal{P}) \to L_4(\mathcal{V}_{\mathbf{G}})$ surjective.

Choose a lift of a class to $L_4(\mathcal{P})$ and take its coordinate classes. A different lift differs by the image of $L_4(\mathcal{S})$, so changes only finitely many coordinates. This proves well-definedness. Every step commutes with the stated functors. □

Notice what the proof does *not* claim: it neither requires nor proves an isomorphism between the L-theory of an infinite product and the product of its L-theories. It uses lifting and coordinate projections only.

Applying ordinary signature to (6.5) and first forgetting labels defines

$$\sigma_{\mathrm{tail}} : L_4(\mathcal{A}_{\mathbf{G}}(X)) \longrightarrow \mathbb{Z}^{\mathbb{N}}/\mathbb{Z}^{(\mathbb{N})}. \tag{6.6}$$

We do not apply this invariant directly to the exterior quotient used in the geometric construction. Localization must first remove the exterior and return a class to an ordinary controlled category.

7 The divisibility obstruction on a fixed free space

7.1 The permutation operation

Let T be nonempty, compact, metrizable, and equipped with a free isometric C_p-action. For each i, let

$$\tau_i = \mathbb{Q}[G_i/P_i]$$

with the orthonormal basis of cosets, its permutation action, and its positive form. Tensoring an object $M_i = \mathbb{Q}[B_i]$ with τ_i gives the basis $(G_i/P_i) \times B_i$, with diagonal G_i-action and label

$$\ell_i(a, b) = \ell_i(b).$$

This basis action is free because the original action on B_i is free. The tensor functor preserves duality and propagation. Define the endomorphism

$$\nu = p\ \mathrm{id} - \tau \otimes (-) \quad \text{on } L_*(\mathcal{A}_{\mathbf{G}}(T)). \tag{7.1}$$

Here p id means addition of p copies of an L-class, not multiplication of every matrix entry by p.

7.2 Local triviality of a finite covering

A *covering map* $\pi : E \to B$ is a map for which every point of B has a neighborhood U whose inverse image is a disjoint union of open sets, each mapped homeomorphically onto U. These open sets are the *sheets*; such a neighborhood is a *trivialization*. For a finite group acting freely on a Hausdorff space, the orbit map is a covering. To prove this, separate the finitely many translates of a point by neighborhoods and shrink one neighborhood so that its nonidentity translates are disjoint.

Thus $T \to T/C_p$ is a p-sheeted covering. Over a trivializing neighborhood there is a continuous *sheet function* c with values in C_p, satisfying

$$c(hx) = \bar{h}\, c(x), \qquad h \in G_i,$$

where $\bar{h}$ is the image in C_p.

Lemma 7.1 (Local identification with p copies)**.** *If a compact set $U \subset T/C_p$ is contained in a covering trivialization, and T_U is its inverse image, then $\nu = 0$ on $L_*(\mathcal{A}_{\mathbf{G},T_U}(T))$.*

Proof. Choose a compact invariant neighborhood L of T_U inside the trivialization. An object supported asymptotically near T_U eventually has all labels in L. The map

$$e_a \otimes e_b \longmapsto e_b \quad \text{in copy } c(\ell_i(b))^{-1}a \in C_p \tag{7.2}$$

is a basis permutation from $\tau_i \otimes M_i$ to p copies of M_i. It is equivariant because the two factors $\bar{h}$ cancel in the copy index. It is orthogonal for the standard forms and has propagation zero.

The sheet pieces of L are disjoint compact sets, hence positively separated. A sequence of matrices with propagation tending to zero eventually has no entries between different sheet pieces. Therefore (7.2) commutes with every controlled morphism, after discarding finitely many indices. It is a natural isometry of functors. Additivity of L-theory gives the assertion. □

7.3 From local equality to global nilpotence

Choose a finite compact cover $U_1, \dots, U_s$ of T/C_p, each contained in a trivialization. Such a cover exists by compactness and by shrinking a finite open cover. Set $T_a = \pi^{-1}(U_a)$. The number s depends on the fixed space T, not on i.

A *nilpotent* endomorphism is one whose some positive power is zero. We claim that ν is nilpotent even though it need not vanish globally.

Lemma 7.2 (Finite-cover nilpotence)**.** *On every L-group of $\mathcal{A}_{\mathbf{G}}(T)$,*

$$\nu^s = 0. \tag{7.3}$$

Proof. Support excision and the localization exact sequence give the *Mayer–Vietoris sequence*

$$\cdots \to L_k(\mathcal{A}_{A\cap B}) \to L_k(\mathcal{A}_A) \oplus L_k(\mathcal{A}_B) \to L_k(\mathcal{A}(T)) \xrightarrow{\partial} L_{k-1}(\mathcal{A}_{A\cap B}) \to \cdots$$

for a closed invariant cover $T = A \cup B$. This sequence expresses how classes on two pieces glue; its connecting map measures the interface obstruction. All its maps commute with ν by naturality.

Induct on the number of trivializing sets, allowing the same assertion for every compact invariant subspace covered by that many sets. The case $s = 1$ is lemma 7.1. For $s > 1$, put

$$A = T_1 \cup \cdots \cup T_{s-1}, \qquad B = T_s.$$

The intersection is contained in the last trivialization, so $\nu = 0$ on its L-groups. For $x \in L_k(\mathcal{A}(T))$,

$$\partial(\nu x) = \nu \partial x = 0.$$

Exactness gives a lift of νx from $L_k(\mathcal{A}_A) \oplus L_k(\mathcal{A}_B)$. The first summand is killed by ν^{s-1} by induction, and the second is killed by ν by local triviality. Since $s - 1 \geq 1$, applying ν^{s-1} to this lift gives $\nu^s x = 0$. □

The nilpotence is a conclusion about an operation on the *controlled* L-groups. The virtual representation $p - \tau_i$ is not being claimed nilpotent in the unrestricted representation ring.

Theorem 7.3 (Integer tail divisibility)**.** *Every $\alpha \in L_4(\mathcal{A}_{\mathbf{G}}(T))$ has a signature tail eventually divisible by p.*

Proof. Extract coordinate Witt classes $[V_i, b_i]$ from α. Naturality of extraction and lemma 7.2 give

$$(p - \tau_i)^s [V_i, b_i] = 0 \quad \text{for all sufficiently large } i. \tag{7.4}$$

A generator g_i of G_i acts on its p quotient cosets by a cycle with no fixed cosets. The character of a permutation representation counts its fixed basis elements, so $\chi_{\tau_i}(g_i) = 0$. Take signature characters in (7.4) and evaluate at g_i:

$$p^s \chi_{\mathrm{Sign}_{G_i}(V_i, b_i)}(g_i) = 0.$$

This is an equality of complex numbers, hence the signature character itself vanishes at g_i. By lemma 6.1, $p \mid \sigma(V_i, b_i)$ on that tail. □

Remark 7.4 (Why rational coefficients do not remove the obstruction)*.* The form $\langle 1/p \rangle$ is a rational form of signature 1, not a form of signature $1/p$. Dividing the coefficients of a positive form does not divide its Witt class additively. The argument uses rational inverses to build matrices, then uses an integer-valued invariant of their classes. Rationalizing the L-groups themselves would be a different operation and would destroy the stated divisibility obstruction.

8 Invariant sphere control and the free detector

Assume now that $\mathbb{Z}_p$ acts faithfully on a connected boundaryless n-manifold, with $n \geq 1$. The purpose of this section is to make all geometric data fixed before choosing the finite quotients.

8.1 A small invariant coordinate neighborhood

Choose a free point by lemma 2.5. Center a chart there and scale its coordinates so that the chart image contains $\overline{B(0,4)} \subset \mathbb{R}^n$. By lemma 2.3, choose $H = H_j$ and $0 < D < 1/64$ such that

$$|hx - x| \leq D \qquad (h \in H,\ x \in \overline{B(0,2)}). \tag{8.1}$$

All points in this formula are expressed in that chart. Shrinking H also ensures that the relevant translates stay in the chart.

Let

$$\Omega = HB(0,2) = \bigcup_{h \in H} hB(0,2).$$

This is the *saturation* of the ball: the smallest H-invariant set containing it. It is open, invariant, and contained in $B(0, 2 + D)$. If $x = hy$ with $y \in B(0,2)$ and $h, k \in H$, then both hy and khy lie within D of y. Therefore

$$|kx - x| \leq 2D \qquad (x \in \Omega,\ k \in H). \tag{8.2}$$

Define the coordinate average

$$F_0(x) = \int_H hx \, d\mu_H(h), \qquad x \in \Omega. \tag{8.3}$$

The integral is a vector in $\mathbb{R}^n$. Translation invariance gives

$$F_0(kx) = \int_H hkx \, d\mu_H(h) = F_0(x).$$

Continuity follows from joint continuity and the parameter estimate for Haar averages. By (8.2),

$$|F_0(x) - x| \le 2D. \tag{8.4}$$

If $x \in \mathrm{Fix}(H)$, every integrand is x, so $F_0(x) = x$.

This averaging does not produce an invariant coordinate chart: F_0 need not be injective, locally invertible, or smooth. We use only its continuity, invariance, closeness to the identity, and behavior at fixed points.

8.2 Pinching a ball onto a sphere

Because the chart center is not fixed by H and $\mathrm{Fix}(H)$ is closed, choose

$$0 < \rho < \rho_+ < 1/4, \qquad \overline{B(0,\rho_+)} \cap \mathrm{Fix}(H) = \varnothing. \tag{8.5}$$

Set

$$Z^+ = F_0^{-1}(\overline{B(0,\rho_+)}). \tag{8.6}$$

It is invariant. Estimate (8.4) places it in $B(0, \rho_+ + 2D)$, whose closure lies in Ω. Since it is closed relative to Ω, it is compact. It contains no H-fixed point: at such a point $F_0(x) = x$, contradicting (8.5).

Choose a continuous *pinch map*

$$s_\rho : \mathbb{R}^n \longrightarrow S^n$$

which maps $B(0,\rho)$ orientation-preservingly and homeomorphically to $S^n \setminus \{\infty\}$, and maps its complement to ∞. For an explicit model, first map the ball to $\mathbb{R}^n$ by $x \mapsto x/(\rho - |x|)$, then use an orientation-compatible inverse stereographic projection. The image tends to ∞ at the ball boundary. The induced map on the compactified source sphere has degree 1.

Define

$$f(x) = s_\rho(F_0(x)) \quad (x \in \Omega), \qquad f(x) = \infty \quad \text{elsewhere in } S^n. \tag{8.7}$$

Its nonconstant locus lies in $B(0, \rho + 2D)$, by (8.4); this is compactly contained in the coordinate region. Hence the extension is continuous. The map is invariant wherever the original action is used.

The formula

$$f_v(x) = s_\rho\big((1-v)F_0(x) + vx\big), \qquad 0 \le v \le 1, \tag{8.8}$$

extended by ∞, is a continuous homotopy from f to s_ρ. Indeed, a nonconstant value forces $|x| < \rho + 2D$, independently of v. Thus f has degree 1. In particular it is not constant, and Z^+ is nonempty.

8.3 The residual free C_p-action

Put $H' = H_{j+1}$. Consider the compact orbit space

$$E = Z^+/H'.$$

A compact group acting continuously on a compact metric space has a compact metrizable Hausdorff orbit space. One concrete description is useful here: replace the metric d on Z^+ by the compatible invariant metric

$$d_H(x, x') = \max_{h \in H} d(hx, hx').$$

Compatibility follows from equicontinuity of the compact family of translations. Then

$$\bar{d}(H'x, H'x') = \min_{h' \in H'} d_H(x, h'x')$$

is a metric on the orbit space, compatible with the quotient topology. Compactness makes the minimum exist; disjoint compact orbits have positive distance.

The group $H/H' = C_p$ acts on E. It acts freely. Indeed, suppose hH' fixes $H'x$. Then $hx = h'x$ for some $h' \in H'$, so $(h')^{-1}h \in H_x$. As $x \notin \mathrm{Fix}(H)$, its closed stabilizer is a proper subgroup of H, hence lies in H' by lemma 2.2. Thus $h \in H'$, proving that the stabilizer in C_p is trivial.

8.4 An explicit finite-dimensional detector

Identify C_p with the pth roots of unity in $S^1 \subset \mathbb{C}$. Denote the quotient character by

$$\chi : H \longrightarrow C_p \subset S^1.$$

The covering $E \to E/C_p$ has a finite trivializing open cover $V_1, \ldots, V_m$. Choose a partition of unity $\lambda_1, \ldots, \lambda_m$ subordinate to it: these are continuous nonnegative functions summing to 1, with $\operatorname{supp} \lambda_j \subset V_j$. For a finite cover of a compact metric space such a partition can be built from distance-to-complement functions after shrinking the cover, and normalizing their sum.

Choose a local section over V_j. Every point above V_j is uniquely a times its section value, with $a \in C_p$. Let c_j denote this sheet coordinate. Define

$$\rho_E(e) = (\sqrt{\lambda_1(\bar{e})}\, c_1(e), \ldots, \sqrt{\lambda_m(\bar{e})}\, c_m(e)) \in \mathbb{C}^m, \tag{8.9}$$

setting the jth component to zero outside the corresponding trivialization. The components are continuous across its boundary because their magnitudes tend to zero there. Their squared magnitudes sum to 1, so the image lies in

$$T = S^{2m-1} \subset \mathbb{C}^m.$$

Moreover $\rho_E(ae) = a\rho_E(e)$. Pulling back along $Z^+ \to E$ gives

$$\widetilde{\rho} : Z^+ \longrightarrow T, \qquad \widetilde{\rho}(hx) = \chi(h)\widetilde{\rho}(x). \tag{8.10}$$

Scalar multiplication by a nonidentity pth root of unity fixes no unit vector, so T is a free C_p-space. Its round metric is invariant. This is a *detector* in the literal sense that it records the residual quotient element as a rotation of a vector. No injectivity or embedding assertion is involved.

8.5 Buffers and the order of choices

Let $y = s_\rho(0)$ and give the target sphere a round metric for which y and ∞ are antipodal. Write $d_\infty = d(y, \infty)$. Choose once and for all

$$0 < r < R < t < u < d_\infty, \qquad Z_0 = \{z : d(y, z) \geq R\}, \qquad Z = T \times Z_0. \tag{8.11}$$

For $v < d_\infty$, set

$$\mathcal{K}_v = f^{-1}(\overline{B(y, v)}).$$

This is compact and lies in the interior of Z^+. To check the latter, note that the inverse image under s_ρ of the compact ball $\overline{B(y, v)}$ is a compact subset of $B(0, \rho)$, which is strictly inside $B(0, \rho_+)$. Thus F_0 has a definite margin to the boundary defining Z^+ on $\mathcal{K}_v$.

Only now choose

$$K_i = H_{\ell_i} \subset H', \quad \ell_i \to \infty, \qquad G_i = H/K_i, \quad P_i = H'/K_i. \tag{8.12}$$

Then $G_i = C_{p^{\ell_i - j}}$ and $G_i/P_i = C_p$. By uniform smallness,

$$\sup_{k \in K_i,\ x \in H\overline{B(0,2)}} |kx - x| \longrightarrow 0.$$

The space T, detector $\widetilde{\rho}$, sets Z^+, Z_0, Z, and all four radii are independent of i. In particular the number of trivializing sets in the nilpotence theorem remains fixed. Choosing a new detector space of increasing complexity at every index would not establish the same fixed-power nilpotence statement.

9 Local chain constructions with shrinking control

The next task is to replace continuous maps by finite rational matrices while retaining uniform support estimates. The repeated mechanism is a finite induction over chain degree. There is no estimate on the size of coefficients or the number of simplices; neither is needed for propagation.

9.1 Acyclic carriers

A *carrier* assigns to each source basis element e a target subcomplex $\Gamma(e)$ in which its image is allowed to lie. It is compatible with boundaries if $\Gamma(e') \subset \Gamma(e)$ whenever e' occurs in de. A map or homotopy is *carried* if the image of each e lies in $\Gamma(e)$.

Lemma 9.1 (The carrier induction)**.** *Let E be a finite based complex in nonnegative degrees, and suppose every $\Gamma(e)$ is augmented acyclic, with compatible augmentations. Two carried chain maps $a, b : E \to F$ that agree in augmentation on degree-zero elements have a carried homotopy $a - b = dH + Hd$. Prescribed compatible degree-zero images also extend to a carried chain map.*

Proof. Assume H has already been defined below the degree of e. The required boundary of $H(e)$ is

$$\gamma_e = a(e) - b(e) - H(de).$$

Its boundary is zero:

$$d\gamma_e = a(de) - b(de) - dH(de) = 0$$

by the induction hypothesis and $d^2e = 0$. In degree zero it also has augmentation zero. Acyclicity therefore supplies a filling $H(e)$ in $\Gamma(e)$. Compatibility with faces ensures all terms already lie there. The chain-map construction uses the same induction, filling the prescribed image of de. □

If every label in $\Gamma(e)$ lies within distance a of the source label, the resulting maps have propagation at most a. After a fixed continuous relabeling c, the bound becomes $\omega_c(a)$.

9.2 Filling without changing the target triangulation

The following refinement is important: all quotient representatives must be maps on the *same* complex at a given index.

Lemma 9.2 (Filling on a fixed target)**.** *Let K triangulate a compact m-manifold and have mesh at most η. Suppose a simplicial r-cycle γ has its whole-simplex support in a set convex in one coordinate chart. Then $\gamma = dq$ for a chain on the same K supported in the closed η-neighborhood of that convex set. For $r = 0$, assume $\varepsilon\gamma = 0$. For $r = m$, the assertion forces $\gamma = 0$.*

Proof. Let A be the subcomplex generated by the support of γ. Contract its inclusion to a point inside the convex set by straight lines in the coordinate chart. This gives a map from the topological cone on A, equal to the inclusion on its base.

Subdivide the cone, not the target K, sufficiently finely to apply simplicial approximation. The simplicial approximation theorem says that a continuous map from a finite complex to $|K|$ has a simplicial approximation after subdivision of its source, chosen using the open-star cover of K [Hat02, Theorem 2C.1]. Let S be the subdivision chain map and a this approximation.

On the base, aS carries every original simplex into itself and fixes vertices. It is therefore exactly the identity on original simplicial chains. Indeed, induct on dimension: the difference on an r-simplex has zero boundary and is an r-chain in that same simplex; there is no nonzero top-dimensional cycle in a simplex when $r > 0$.

Let c be the augmented cone operator, so $dc + cd = 1$. Then

$$q = aS(c\gamma), \qquad dq = aS\gamma = \gamma.$$

For the support estimate, the open-star condition places every image simplex in a target simplex meeting the image of the cone map, hence meeting the convex set. Its diameter is at most η. If $r = m$, the target has no $(m+1)$-chains, so $q = 0$ and $\gamma = 0$. □

The use of whole-simplex support prevents a hidden error: controlling only the barycenters of chains before taking faces or fillings is not sufficient. A mesh estimate converts whole-simplex support bounds into label bounds afterwards.

9.3 Dual blocks and controlled cap inverses

Let K be a combinatorial triangulation of an oriented compact m-manifold A, with boundary a subcomplex. The *dual block* $D\sigma$ to a simplex σ consists of the barycentric simplices whose flags begin with σ, together with their faces. For an interior simplex it is a ball of dimension $m - \dim\sigma$. With suitable orientations,

$$\partial D\sigma = (-1)^{\dim\sigma+1} \sum_{\substack{\tau\supset\sigma \\ \dim\tau=\dim\sigma+1}} [\tau : \sigma] D\tau, \tag{9.1}$$

where $[\tau : \sigma]$ is the coefficient of σ in $d\tau$. For a top simplex ρ, its dual is the signed vertex $D\rho = \varepsilon_\rho[b_\rho]$, where ε_ρ is its fundamental coefficient.

Let D_{rel} be the cellular complex generated by blocks of simplices not contained in ∂A. All cofaces of such a simplex are again not contained in the boundary, so these blocks form a subcomplex. Set

$$J_\sigma = \bigcup_{\substack{\tau \supseteq \sigma \\ \tau \not\subseteq \partial A}} D\tau.$$

For an interior σ, this is $D\sigma$, hence a ball. For a boundary simplex, its link is a ball and its boundary is the link in ∂A. The displayed incidence formula identifies $C_*(J_\sigma)$, with reversed degrees and orientation signs, with the relative cochain complex of that link and its boundary. A ball relative to its boundary has reduced homology concentrated in the top degree, with one generator; thus J_σ is augmented acyclic. The augmentation is nonzero on every signed top-coface dual vertex and identifies H_0 with $\mathbb{Q}$.

Lemma 9.3 (Local dual-block comparison)**.** *There are chain maps*

$$S_{\mathrm{rel}} : D_{\mathrm{rel}} \longrightarrow C_*(\mathrm{sd}\, K), \qquad R_{\mathrm{rel}} : C_*(\mathrm{sd}\, K) \longrightarrow D_{\mathrm{rel}},$$

with $R_{\mathrm{rel}}S_{\mathrm{rel}} = 1$ and $S_{\mathrm{rel}}R_{\mathrm{rel}} \simeq 1$, all with propagation at most $C_m\eta$ when cells are labeled by points of their supports.

Proof. Take S_{rel} to be cellular subdivision. For a barycentric flag $\sigma_0 < \cdots < \sigma_k$, use J_{σ_0} as the carrier for R_{rel}. Deleting the minimum face replaces it by a smaller carrier; deleting another face leaves the carrier unchanged. The carrier induction therefore constructs the map, with augmentation-preserving values on vertices.

On each positive-dimensional dual block the composite $R_{\mathrm{rel}}S_{\mathrm{rel}}$ equals the identity by induction on dimension: its difference has zero boundary and is a top-dimensional cycle in that dual ball, hence is zero. The degree-zero case follows from augmentation. For the other composite, both maps are carried in the subdivided closed star of the minimum face, which is contractible, so lemma 9.1 gives a homotopy.

Each such carrier lies in a closed star of diameter at most 2η. A fixed number of compositions gives the asserted bound. It depends on dimension, not on how many simplices meet a face. □

Let $R = C_*(K, \partial K; \mathbb{Q})$. The map

$$\mathcal{J}_r : R_r^{m-*} \longrightarrow (D_{\mathrm{rel}})_r, \qquad \mathcal{J}_r(\sigma^*) = (-1)^{r(m+1)} D\sigma \tag{9.2}$$

is a chain isomorphism by (9.1) and the differential in (5.1).

Lemma 9.4 (A cap map with a controlled inverse)**.** *Suppose $F : R^{m-*} \to C_*(K)$ is carried on σ^* in $\mathrm{st}_K\, \sigma$, and on top simplices satisfies $\varepsilon F(\rho^*) = \varepsilon_\rho$. Then F is a chain homotopy equivalence with inverse and both inverse homotopies of propagation $O_m(\eta)$.*

Proof. Let $S_K : C_*(K) \to C_*(\mathrm{sd}\, K)$ be subdivision. A simplex-carried inverse R_K and its homotopies are constructed by the carrier lemma, with $R_K S_K = 1$ by the same top-chain uniqueness argument as above. Put

$$F_0 = R_K S_{\mathrm{rel}} \mathcal{J}, \qquad B = \mathcal{J}^{-1} R_{\mathrm{rel}} S_K.$$

The preceding identities give $F_0 B \simeq 1$ and $B F_0 \simeq 1$ through local homotopies.

Both F and F_0 have the prescribed degree-zero augmentations and are carried in closed stars. The shifted dual differential takes a simplex to cofaces, and $\mathrm{st}_K \tau \subset \mathrm{st}_K \sigma$ when $\tau \supset \sigma$. Thus the carrier lemma gives $F - F_0 = dH + H\delta$ in the same carriers. Adding HB and BH to the two inverse homotopies shows that B is also a controlled inverse for F. Only finitely many local compositions occur. □

Apply this to the slant of $t_A = \mathrm{Sym}\, \Delta z_A$. On relative cochains its boundary term disappears, because

$$dt_A = j^{\otimes 2} t_{\partial A}. \tag{9.3}$$

On a top simplex $\rho = [v_0, \dots, v_m]$ the degree-zero slant is $\varepsilon_\rho([v_0] + [v_m])/2$, of augmentation ε_ρ. Hence the lemma supplies the controlled cap inverse. Using the *same ordered diagonal* on A and its boundary makes (9.3) exact, with no change of boundary structure.

9.4 The common closed complexes

Fix a combinatorial triangulation of $W = S^n \times \mathbb{CP}^2$ compatible with a finite PL disk and equator flag to be used for the final cuts. Such a triangulation can be obtained by triangulating the pieces and their products compatibly. Let Y_i be successive refinements with mesh $\eta_i \to 0$, and put

$$D_i = C_*(Y_i; \mathbb{Q}), \qquad z_i = z_{W,i}.$$

Label each basis simplex at its barycenter. The preceding construction, with empty boundary, gives strict symmetric equivalences

$$\phi_i : D_i^{N-*} \longrightarrow D_i, \qquad b_i : D_i \longrightarrow D_i^{N-*}, \tag{9.4}$$

with both inverse homotopies of propagation $O_N(\eta_i)$ in the metric of W. A fixed continuous relabeling converts these to asymptotically small matrices.

A compact equicontinuous family of self-maps of W has chain approximations on the unchanged D_i. Subdivide each source enough for the target open-star cover, apply simplicial approximation, and compose its chain map with the subdivision map. If ω is a common modulus of continuity, the whole-simplex graph error is bounded by

$$\mu_i = O(\eta_i + \omega(C\eta_i)) \longrightarrow 0. \tag{9.5}$$

This means that an image simplex is within this error of the image of its original source simplex. Extra source subdivisions may depend on the map; after composition, all matrices still have source and target D_i. The identity is taken to have chain approximation exactly 1.

10 Representatives of the finite quotients on chains

10.1 Completing local homeomorphisms to sphere maps

Choose a continuous cutoff function θ equal to 1 on $B(0,1)$ and to 0 outside $B(0,3/2)$. For $h \in H$, define on the coordinate sphere

$$\widehat{h}(x) = x + \theta(x)(hx - x), \tag{10.1}$$

and use the identity where the displayed coordinate formula is not needed. This is a continuous sphere map, not necessarily a homeomorphism. Replacing θ by $v\theta$ gives a homotopy to the identity, so it has degree 1. The family is equicontinuous because the original action is jointly continuous on a compact parameter set and a compact coordinate region.

Crucially,

$$f\widehat{h} = f. \tag{10.2}$$

Inside $B(0,1)$ this follows from H-invariance of F_0. Outside that ball, both the original point and its image remain away from $B(0,\rho+2D)$, the possible nonconstant locus of f.

Choose representatives $h_g \in H$ for $g \in G_i$, with $h_1 = 1$. In multiplicative notation for the action,

$$h_g h_h = h_{gh} k_{g,h}, \qquad k_{g,h} \in K_i.$$

Although $\mathbb{Z}_p$ is an additive group, this notation records composition of its acting homeomorphisms. Choose chain approximations

$$a_{g,i} : D_i \longrightarrow D_i$$

to $\widehat{h}_g \times 1_{\mathbb{CP}^2}$, with $a_{1,i} = 1$. They have a common graph-error bound $\mu_i \to 0$ from (9.5).

Each map has degree 1 on W, so it induces the identity on the fundamental homology class. Because D_i has no chains in degree $N+1$, homologous top cycles are equal as chains. Therefore

$$a_{g,i} z_i = z_i. \tag{10.3}$$

10.2 Graph control

A matrix is not required to move points a small distance in the original coordinate chart. A representative h_g can have a fixed small but nonvanishing displacement. The appropriate estimate instead remembers the quotient element it represents.

Say that a family of matrices has *graph type* g if its propagation in the f-coordinate tends to zero and, for every fixed $v < d_\infty$, its entries with source first coordinate $x \in \mathcal{K}_v$ satisfy

$$d_T(\widetilde{\rho}(x'), \chi(g)\widetilde{\rho}(x)) \longrightarrow 0 \tag{10.4}$$

uniformly, where x' is the target first coordinate. Target detector labels exist eventually: the small f-propagation places targets in a slightly larger fixed $\mathcal{K}_{v'}$, with $v < v' < d_\infty$.

The notation $\chi(g)$ is well-defined because $K_i \subset H'$ lies in the kernel of χ. Equation (10.2), the chain graph estimate, and detector equivariance show that $a_{g,i}$ has type g.

Graph types compose. For example, an entry following type h and then type g has detector label close to $\chi(g)\chi(h)\widetilde{\rho}(x) = \chi(gh)\widetilde{\rho}(x)$. Use a larger fixed buffer for the intermediate label. Transposition reverses type: if the target lies in $\mathcal{K}_v$, small f-propagation first places the original source in $\mathcal{K}_{v'}$, and (10.4) then gives type g^{-1} for the transpose. The same statements hold for chain homotopies, which are matrices of degree 1.

10.3 Product homotopies on the unchanged complexes

In the finite coordinate region consider the tracks

$$L_{g,h}(x,v) = (1-v)\widehat{h}_g\widehat{h}_h(x) + v\widehat{h}_{gh}(x), \qquad 0 \le v \le 1. \tag{10.5}$$

They extend by the identity track near ∞. For $|x| \le 1/2$, the endpoint maps agree with the original action where they are evaluated. Their difference is caused by $k_{g,h} \in K_i$, and uniform smallness plus equicontinuity gives a bound $\beta_i \to 0$ from the entire track to $h_{gh}x$. For $|x| \ge 1/2$, the entire track lies within $2D$ of x and has f-value ∞.

Lemma 10.1 (Uniform product homotopies)**.** *There are homotopies on the same D_i*

$$a_g a_h - a_{gh} = dB_{g,h} + B_{g,h}d \tag{10.6}$$

of graph type gh, with error bounds uniform in g, h and tending to zero with i.

Proof. We give the support induction, since merely knowing that the maps are homotopic would not give the required control. Assume the values of B on proper faces of a simplex σ have been chosen. The residual

$$\gamma_\sigma = a_g a_h(\sigma) - a_{gh}(\sigma) - B(d\sigma)$$

is a cycle, and in degree zero has augmentation zero. Choose a product coordinate chart containing the track of σ in (10.5) and its unchanged second-coordinate projection. In the first coordinate take the convex hull of the track; in the second take a small coordinate-convex carrier. Include the whole endpoint-chain supports and the already chosen face fillings, then take their coordinate convex hull. Fill γ_σ on the original target triangulation using lemma 9.2.

Here is the uniform estimate in that induction. Fix finite chart buffers and metric comparison constants on W. Put

$$\delta_i = \eta_i + \mu_i + \omega(C\eta_i) + \omega(C\mu_i) + \beta_i \longrightarrow 0.$$

If the fillings through degree $r-1$ stay within width w_{r-1} of their nested track hulls, then including those fillings, changing product coordinates, and adding one target mesh gives

$$w_{-1} = 0, \qquad w_r \le C(w_{r-1} + \delta_i). \tag{10.7}$$

The coordinate hull for a face is contained in that for its parent before the added widths. Thus this recurrence applies at each step. Only $N+1$ steps occur, and N is fixed, so $w_r \to 0$ uniformly for all relevant degrees. In the region where the tracks have a fixed displacement at most $2D$, the first factor remains in the fixed Euclidean chart; the second-factor carriers shrink to points. Near infinity all maps are the identity and small sphere charts suffice. At top degree the residual is zero by the final assertion of lemma 9.2; no extra chain degree is introduced.

For a source barycenter with $|x| \le 3/8$, the supports lie within $o(1)$ of $h_{gh}x$, proving the detector part of type gh on every fixed buffer. For $|x| \ge 3/8$, they lie within $2D + o(1)$ of x. The fixed inequality

$$3/8 - \rho - 4D > 1/16$$

keeps them outside the nonconstant locus of f. Hence all these entries have f-value ∞ at both ends. Uniform continuity gives the asserted global f-control. □

10.4 Compatibility with the Poincaré structure

We also require controlled homotopies expressing preservation of duality. For a fixed g, consider

$$a_g \phi a_g^* \quad \text{and} \quad \phi : D^{N-*} \longrightarrow D.$$

Both are geometrically local with error tending to zero. For the first composite, an entry of a_g^* selects a source cell mapping near the given target cell. The cap map ϕ moves only $O(\eta_i)$, and the final a_g maps the nearby cell back near that same target, by equicontinuity. This argument does not require $\widehat{h}_g$ to be injective.

Their degree-zero augmentations agree. For $\alpha \in D_N^*$,

$$\varepsilon\phi(\alpha) = \alpha(z_i), \qquad \varepsilon a_g \phi a_g^*(\alpha) = \alpha(a_g z_i) = \alpha(z_i).$$

Apply the filling induction to the shifted dual source. Its differential goes from a simplex to cofaces, so take coordinate-convex carriers containing the endpoint supports and the carriers already assigned to those cofaces. Every coface meets the original simplex. The same finite-degree width estimate keeps the fillings within $o(1)$ of it. We obtain a local primitive

$$a_g \phi a_g^* - \phi = dU_g + U_g\delta. \tag{10.8}$$

In particular, U_g has graph type 1.

Put $a = a_g$, $b' = a_{g^{-1}}$. Let T be the homotopy $b'a - 1 = dT + Td$ from lemma 10.1. Its shifted dual primitive is

$$K_r = (-1)^{r+1} T^*_{N-r-1}, \qquad \delta K + K\delta = a^*(b')^* - 1.$$

Using

$$a\phi - \phi(b')^* = a\phi(1 - a^*(b')^*) + (a\phi a^* - \phi)(b')^*,$$

we get

$$V_g = -a\phi K + U_g(b')^*, \qquad dV_g + V_g\delta = a_g\phi - \phi a^*_{g^{-1}}. \tag{10.9}$$

The maps T, K, U_g have type 1, while $(b')^*$ has type g, so V_g has type g. All errors are uniform over the finite quotients because the estimates used the original compact family H, not an index-dependent modulus.

Fix one null sequence ε_i bounding the f-errors, detector errors on the selected buffers, local duality propagations, and all primitive homotopies just constructed. Enlarging a null sequence by a fixed finite maximum is harmless.

11 Compression to the detector region

The detector is defined only on Z^+. We therefore cannot simply label every simplex of W by $(\widetilde{\rho}, f)$. Compression removes the chains far from the region of interest; the resulting defects are confined to an exterior band.

11.1 The quotient complex and its based section

Let $F_i \subset D_i$ be generated by those simplices whose whole closed simplex misses $f^{-1}(\overline{B(y,t)}) \times \mathbb{CP}^2$. It is a subcomplex: every face of a missing simplex also misses that set. Put

$$C_i = D_i/F_i, \qquad \pi : D_i \to C_i, \qquad \sigma : C_i \to D_i, \qquad Q = 1 - \sigma\pi. \tag{11.1}$$

The map σ is the degreewise based section retaining the remaining simplices. It is generally *not* a chain map. The quotient differential is

$$\widehat{d} = \pi d\sigma.$$

Since F_i is a subcomplex, π is a chain map and $\pi dQ = 0$. The projection Q selects the discarded, or *far*, basis elements.

Let $\kappa_i \to 0$ bound the oscillation of f on a simplex. A retained simplex meets the target t-ball, so its barycenter label has target radius at most $t + \kappa_i$. A far simplex has barycenter radius greater than t. Since $t < u$, all retained detector labels are defined for large i.

11.2 Why all compression errors are exterior errors

A *word* in the primitive matrices means a composable product of a fixed number of differentials, cap maps, inverses, representative maps, homotopies, their duals, or the based sections and projections. Its length is the number of factors; it is bounded independently of i.

Lemma 11.1 (Exterior-band factorization)**.** *A retained-to-retained word of length at most L containing Q or Q^* factors through a based module with labels in the band*

$$t - L\varepsilon_i \le d(y, f(x)) \le t + \kappa_i + L\varepsilon_i. \tag{11.2}$$

For local words, the factors are controlled in the scalar detector-and-sphere coordinates. For graph words, the factors become equivariant controlled maps after the sheet compensation described below. The intermediate object is eventually supported in the exterior $Z = T \times Z_0$.

Proof. At one occurrence of Q, retain only far basis elements reached from a retained source and contributing to a retained target. Cutting the product there gives an exact factorization of matrices. Each contributing path has at most L steps, each with f-error at most ε_i. The retained endpoints have radius at most $t + \kappa_i$, and the far cut has radius greater than t. These facts give (11.2), including the corresponding bounds on other contributing intermediate labels.

The upper bound is eventually less than u, so every intermediate detector label needed for the factorization is defined. The lower bound is eventually greater than R, because $t - R > 0$ is fixed. Thus the cut object lies in the exterior.

For graph type g, introduce a sheet coordinate $s \in G_i$, give (x, s) the label

$$(\chi(s)\widetilde{\rho}(x), f(x)), \tag{11.3}$$

and accompany the graph map by the sheet translation $s \mapsto sg^{-1}$. Then its target detector label is close to

$$\chi(sg^{-1})\chi(g)\widetilde{\rho}(x) = \chi(s)\widetilde{\rho}(x).$$

The detector displacement cancels. Use all sheets for the cut object, so the basis action is free and the factorization equivariant. Apply this argument separately to the prefix and suffix of the word. A sum of words factors through the direct sum of their cut objects; its propagation is bounded by the maximum of the summand bounds. □

The last sentence is important for averages over G_i: the number of summands can grow with i, because ranks are unrestricted. The required bounds are uniform bounds on their propagation, not bounds on their number.

11.3 Compressed maps and exact defect formulas

Define

$$A_g = \pi a_g \sigma, \qquad \widehat{\phi} = \pi\phi\pi^*, \qquad \widehat{b} = \sigma^* b \sigma. \tag{11.4}$$

The maps A_g are compressed representatives, not yet an honest action by chain maps on the unquotiented detector category. Every failure is accounted for by inserting $1 - Q = \sigma\pi$.

For raw operators α, β with compatible degrees, write $\widehat{\alpha} = \pi\alpha\sigma$. Then

$$\widehat{\alpha}\widehat{\beta} - \widehat{\alpha\beta} = -\pi\alpha Q\beta\sigma, \tag{11.5}$$

$$\widehat{d}A_g - A_g\widehat{d} = \pi a_g Q d\sigma, \tag{11.6}$$

$$A_g A_h - A_{gh} - \widehat{d}\widehat{B}_{g,h} - \widehat{B}_{g,h}\widehat{d} = -\pi a_g Q a_h \sigma + \pi B_{g,h} Q d\sigma, \tag{11.7}$$

where $\widehat{B}_{g,h} = \pi B_{g,h}\sigma$. Each term on the right contains Q and has bounded word length, so the band lemma applies.

The map $\widehat{\phi}$ is an actual strict symmetric chain map because π is a chain map. To verify that it is Poincaré in the exterior quotient, put

$$e = d\sigma - \sigma\widehat{d} = Qd\sigma, \qquad e^\vee = \widehat{\delta}\sigma^* - \sigma^*\delta.$$

The second map is the signed shifted dual of e, so it factors through Q^*. A direct computation gives

$$\widehat{\delta}\widehat{b} - \widehat{b}\widehat{d} = e^\vee b\sigma + \sigma^* b e. \tag{11.8}$$

If $\phi b - 1 = dH + Hd$ and $b\phi - 1 = \delta H' + H'\delta$, then

$$\widehat{\phi}\widehat{b} - 1 = \widehat{d}(\pi H\sigma) + (\pi H\sigma)\widehat{d} + \pi H e - \pi\phi Q^* b\sigma, \tag{11.9}$$

$$\widehat{b}\widehat{\phi} - 1 = \widehat{\delta}(\sigma^* H'\pi^*) + (\sigma^* H'\pi^*)\widehat{\delta} - e^\vee H'\pi^* - \sigma^* b Q\phi\pi^*. \tag{11.10}$$

All residual terms are local words covered by lemma 11.1. Thus $\widehat{b}$ becomes a chain inverse for $\widehat{\phi}$ in $\mathcal{B}_{\mathbf{1},Z}(T \times S^n)$.

Similarly, with $\widehat{V}_g = \pi V_g \pi^*$,

$$\begin{aligned} A_g\widehat{\phi} - \widehat{\phi}A^*_{g^{-1}} - \widehat{d}\widehat{V}_g - \widehat{V}_g\widehat{\delta} \\ = -\pi a_g Q\phi\pi^* + \pi\phi Q^* a^*_{g^{-1}}\pi^*. \end{aligned} \tag{11.11}$$

These are graph-type g residuals, hence vanish in the equivariant exterior quotient after compensation. Also $A_1 = \pi\sigma = 1$ exactly. For completeness, the compressed fundamental chain obeys

$$A_g \pi z_i - \pi z_i = -\pi a_g Q z_i;$$

the residual is supported outside the retained inner R-ball eventually.

We have obtained a Poincaré complex $(C, \widehat{\phi}/2)$ in the scalar exterior category with detector labels, together with group-law and adjoint homotopies whose compensated versions are controlled in the equivariant exterior category. These are precise quotient statements; none asserts that Q is small as a raw matrix.

11.4 The scalar exterior class

For a continuous labeling map $c : W \to S^n$, define

$$\mathfrak{m}_c = [(D_i, \phi_i/2) \text{ labeled by } c] \in L_N(\mathcal{B}_{\mathbf{1},Z_0}(S^n)).$$

The class is well-defined because a fixed continuous label map takes the local cap data to asymptotically controlled data.

After forgetting T, all of D_i can be labeled by f. The discarded subcomplex F_i is Z_0-supported, since its labels have radius greater than $t > R$. In that scalar exterior quotient, Q and e vanish, so π and σ are inverse chain equivalences. Transporting the original structure along π gives exactly $\widehat{\phi}/2$. Therefore

$$[(C_i, \widehat{\phi}_i/2) \text{ with sphere labels}] = \mathfrak{m}_f. \tag{11.12}$$

12 The free-sheet idempotent and its normalized corner

12.1 Why sheets restore control

Form

$$\mathcal{C}_i = C_i \otimes \mathbb{Q}[G_i].$$

The strict G_i-action is on the sheet factor only: $h(c \otimes s) = c \otimes hs$. If x_c is the first coordinate of the barycenter of c, label the basis by

$$\ell(c \otimes s) = (\chi(s)\widetilde{\rho}(x_c), f(x_c)) \in T \times S^n. \tag{12.1}$$

This is an equivariant labeling of a free G_i-basis. Copying the local compressed differential and duality on every sheet gives an equivariant Poincaré complex in the exterior quotient, with

$$\Phi = \widehat{\phi} \otimes 1.$$

For $g \in G_i$, let $R_{g^{-1}}$ send sheet s to sg^{-1}. Right sheet translation commutes with the strict left sheet action. By the cancellation in lemma 11.1, $A_g \otimes R_{g^{-1}}$ is controlled even though A_g by itself has graph type g, rather than small unshifted detector propagation.

Put

$$q_i = |G_i|, \qquad E_i = \frac{1}{q_i} \sum_{g \in G_i} A_{g,i} \otimes R_{g^{-1}}. \tag{12.2}$$

All entries are rational. The sequence $q = (q_i)$ defines an invertible central scalar morphism: it commutes with every matrix, equals its own transpose, and both q and q^{-1} have propagation zero. Its size is unrestricted.

12.2 The homotopy idempotent equations

The matrices E_i satisfy, in the equivariant exterior homotopy category,

$$E^2 \simeq E, \qquad E\Phi \simeq \Phi E^*. \tag{12.3}$$

For the first equation, expand

$$E^2 - E = \frac{1}{q_i^2} \sum_{g,h} (A_g A_h - A_{gh}) \otimes R_{(gh)^{-1}}.$$

Each $k \in G_i$ occurs as gh in exactly q_i pairs, which accounts for the subtracted E. The compensated sum

$$\frac{1}{q_i^2} \sum_{g,h} \widehat{B}_{g,h} \otimes R_{(gh)^{-1}}$$

is a controlled homotopy, up to the exterior witnesses in (11.7). The sum of those witnesses factors through their direct sum, still an allowed object.

For the second equation, use $R_{g^{-1}}^* = R_g$ and reindex by inversion:

$$E\Phi - \Phi E^* = \frac{1}{q_i} \sum_{g} (A_g \widehat{\phi} - \widehat{\phi} A_{g^{-1}}^*) \otimes R_{g^{-1}}.$$

Equation (11.11) and the compensated $\widehat{V}_g$ give the required homotopy and exterior factorizations. The chain-map defect of E is handled by (11.6) in the same way.

Thus E is not an approximately idempotent matrix in an operator norm. It is an actual idempotent *in the homotopy category of the exterior quotient*, because the displayed defects are chain homotopies there.

12.3 Splitting a duality-compatible homotopy idempotent

We record the algebraic calculation in a general rational additive category $\mathcal{Q}$ with involution. Rational means that morphism groups are $\mathbb{Q}$-vector spaces and composition is $\mathbb{Q}$-bilinear.

Lemma 12.1 (The normalized Poincaré corner)**.** *Suppose $\mathcal{C} \in [0, N]$ has a strict symmetric Poincaré equivalence Φ, and E is a chain map satisfying (12.3). Let q be an invertible central self-adjoint scalar. There is a splitting*

$$D \xrightarrow{\iota} \mathcal{C} \xrightarrow{r} D, \qquad \iota r \simeq E, \quad r\iota \simeq 1, \tag{12.4}$$

in $K^b(\operatorname{Kar} \mathcal{Q})$, with $D \in [0, N]$. It carries the strict symmetric Poincaré structure

$$\phi_D = q^{-1} r \Phi r^*, \qquad b_D = q\, \iota^* \Phi^{-1} \iota \tag{12.5}$$

as a homotopy inverse. The quadratic structure $\phi_D/2$ defines a class in $L_N(\mathcal{Q})$ independent of the chosen splitting.

Proof. Apply foundational theorem 5.2 to $[E]$ in $K^b(\operatorname{Kar} \mathcal{Q})$. The splitting initially gives a bounded complex. If it has terms below 0, then on its lowest nonzero degree the homotopy $r\iota \simeq 1$ expresses the identity as dH because $\mathcal{C}$ is zero there. Thus the extremal differential is a split surjection. Split off that contractible two-term summand. At a degree above N, the dual argument gives a split injection and removes the upper contractible summand. Repeating finitely many times puts D in $[0, N]$.

Conjugating a strictly symmetric map by r preserves strict symmetry. For the inverse equations, use $\iota r \simeq E$ and the compatibility in (12.3):

$$\phi_D b_D \simeq r\Phi E^*\Phi^{-1}\iota \simeq rE\iota \simeq 1,$$
$$b_D\phi_D \simeq \iota^*\Phi^{-1}E\Phi r^* \simeq \iota^*E^*r^* \simeq 1.$$

The factor q cancels q^{-1}, and centrality permits the indicated rearrangements.

For two splittings, $r_2\iota_1$ is a chain equivalence between the resulting complexes. The same equations show that it carries the first symmetric structure to the second up to homotopy. The rational primitive argument identifies the quadratic structures, and the Poincaré mapping cylinder gives equality of their ordinary classes in the completed category. Finally use the natural ultimate comparison $L_N(\mathcal{Q}) \cong L_N(\operatorname{Kar}\mathcal{Q})$ from foundational theorem 5.1. □

The name *corner* means the part of the Poincaré complex selected by the idempotent. It is analogous to restricting a bilinear form to an orthogonally split summand, except that both the projection and the orthogonality equations hold up to chain homotopy.

For the application, take

$$\mathcal{Q} = \mathcal{B}_{\mathbf{G},Z}(T \times S^n).$$

The splitting is made *once in the category of sequences*. Consequently its finitely many maps and homotopies already have asymptotically small representatives. Choosing an unrelated splitting at every index would not by itself give this control. We obtain the class

$$\mathfrak{d} = [D, \phi_D/2] \in L_{n+4}(\mathcal{B}_{\mathbf{G},Z}(T \times S^n)). \tag{12.6}$$

12.4 The normalization calculation after forgetting

Let U forget both equivariance and the T-coordinate, retaining sphere control. All the A_g and their relation homotopies are uniformly controlled in this remaining coordinate. This permits the following sheet maps.

In sheet coordinates, (12.2) has block entries

$$E_{ab} = q^{-1}A_{a^{-1}b}. \tag{12.7}$$

Let $j_1 : C \to \mathcal{C}$ include the identity sheet and $\pi_1 : \mathcal{C} \to C$ project to it. Define

$$v = Ej_1, \qquad u = q\pi_1 E, \qquad v_a = q^{-1}A_{a^{-1}}, \quad u_b = A_b. \tag{12.8}$$

The group-law homotopies give

$$uv \simeq 1, \quad vu \simeq E, \quad uE \simeq u, \quad Ev \simeq v. \tag{12.9}$$

For example, $uv = q^{-1}\sum_b A_bA_{b^{-1}} \simeq 1$, and the (a,b) block of vu is $q^{-1}A_{a^{-1}}A_b \simeq E_{ab}$.

The adjoint homotopies imply $A_b\widehat{\phi}A_b^* \simeq \widehat{\phi}$, hence

$$u\Phi u^* = \sum_b A_b\widehat{\phi}A_b^* \simeq q\widehat{\phi}. \tag{12.10}$$

Now $w = u\iota : D \to C$ and $z = rv : C \to D$ are inverse chain equivalences, and

$$\begin{aligned} w\phi_D w^* &\simeq q^{-1} u E \Phi E^* u^* \\ &\simeq q^{-1} u \Phi u^* \\ &\simeq \widehat{\phi}. \end{aligned} \tag{12.11}$$

A chain equivalence satisfying this equation is a *homotopy isometry*: it preserves the Poincaré form up to the prescribed chain homotopy. The rational half-unit comparison makes it an identification of quadratic classes as well.

Every sum in (12.8)–(12.10) has uniform propagation bounds. Its number of terms and coefficients may grow with i, but the category permits this. Equations (11.12) and (12.11) give the central scalar comparison

$$U_* \mathfrak{d} = \mathfrak{m}_f. \tag{12.12}$$

The factor q^{-1} in (12.5) is exactly what cancels the q in (12.10). Replacing this calculation by an informal assertion that one has taken the invariant part would lose the required normalization.

13 Localization boundaries and their geometric meaning

13.1 The boundary in an exterior quotient

For a closed invariant cover $X = A \cup B$, put $Y = A \cap B$. The ordinary support Mayer–Vietoris boundary has degree -1:

$$\partial_{A,B} : L_k(\mathcal{A}(X)) \longrightarrow L_{k-1}(\mathcal{A}_Y(X)).$$

We need the same boundary when an exterior $Z \subset A$ has already been discarded.

Proposition 13.1 (Exterior localization boundary)**.** *Suppose $Z \subset A$ is closed invariant and $d(Z, B) > 0$. There is a natural map*

$$\partial_{A,B} : L_k(\mathcal{A}(X)/\mathcal{A}_Z(X)) \longrightarrow L_{k-1}(\mathcal{A}_Y(X)). \tag{13.1}$$

It commutes with duality-preserving functors that preserve the specified support subcategories.

Proof. Put

$$\mathcal{Q} = \mathcal{A}(X)/\mathcal{A}_Z, \qquad \mathcal{U} = \mathcal{A}_A/\mathcal{A}_Z.$$

Positive separation embeds $\mathcal{A}_B$ and $\mathcal{A}_Y$ fully faithfully in $\mathcal{Q}$. The based support decompositions used in proposition 4.3 still give filtrations after quotienting: choose representatives for the finite collection of quotient maps and make all their needed decompositions before taking the quotient.

Support excision gives

$$\mathcal{U}/\mathcal{A}_Y \simeq \mathcal{Q}/\mathcal{A}_B. \tag{13.2}$$

On A-supported objects, the ideal through B restricts to the ideal through Y. The ideal through Z already factors inside the A-supported category. Thus the two quotient descriptions use, respectively, the ideals $I_Z + I_Y$ and $I_Z + I_B$, and the same based split proves the equivalence.

At the level of L-groups, take the composite

$$L_k(\mathcal{Q}) \longrightarrow L_k(\mathcal{Q}/\mathcal{A}_B) \cong L_k(\mathcal{U}/\mathcal{A}_Y) \xrightarrow{\partial} L_{k-1}(\mathcal{A}_Y).$$

The last arrow is the localization connecting homomorphism. Naturality follows from that of the quotient functors and localization. □

13.2 The boundary of a union is the interface

Lemma 13.2 (Union-boundary rule)**.** *Let $V_A = (P \to D_A)$ and $V_B = (P \to D_B)$ be k-dimensional Poincaré pairs supported on A and B, with oppositely oriented common boundary P supported on Y. Orient P as the boundary of V_A. In the ordinary or the separated exterior setting,*

$$\partial_{A,B}[V_A \cup_P V_B] = [P]. \tag{13.3}$$

Proof. In the relative-pair description of localization, the pair $P \to D_A$ represents a relative class whose connecting image is $[P]$. In the quotient by the B-supported category, the entire B-piece vanishes, as do the interface and its suspended copies in the algebraic union. The surviving complex and induced quadratic structure are precisely the A-piece. Excision (13.2) therefore identifies the quotient image of the union with the relative-pair class. Apply the connecting map. This is the pair-to-quotient comparison in [CP95, proof of Theorem 4.1], with the natural passage to ultimate decoration. □

The rule is not merely a statement about underlying chain complexes. The induced quadratic structures must agree. The following two sections give the relative duality and union tensors that make this verification explicit for our manifold models.

14 Relative duality and controlled manifold unions

14.1 The relative cap equivalence

Let A be an oriented compact combinatorial m-manifold with boundary. Write

$$P = C_*(\partial A; \mathbb{Q}), \quad D = C_*(A; \mathbb{Q}), \quad j : P \longrightarrow D, \quad J = \mathrm{Cone}(j).$$

Use one triangulation, one vertex order, and the diagonal induced by that order on all subcomplexes. Fundamental chains give

$$t_D = \mathrm{Sym}\,\Delta z_A, \qquad t_P = \mathrm{Sym}\,\Delta z_{\partial A}, \qquad dt_D = j^{\otimes 2} t_P, \quad dt_P = 0.$$

Let ϕ_D, ϕ_P be their slants. The relative Poincaré map is

$$\Psi_r(\alpha, \beta) = \phi_{D,r}\alpha + (-1)^{r+1} j\phi_{P,r}\beta : J_r^{m-*} \longrightarrow D_r. \tag{14.1}$$

Its domain is $D^*_{m-r} \oplus P^*_{m-r-1}$. From (3.2), its differential is

$$d_{J^{m-*},r} = (-1)^r \begin{pmatrix} d_D^* & 0 \\ j^* & -d_P^* \end{pmatrix}.$$

The tensor boundary equation gives

$$d\phi_{D,r} - (-1)^r \phi_{D,r-1} d^* = j\phi_{P,r-1} j^*,$$

which cancels exactly with the extra term in (14.1). Thus Ψ is a chain map.

Lemma 14.1 (Controlled relative Poincaré duality)**.** *For mesh η, the map Ψ, a chain inverse, and both inverse homotopies can be chosen with propagation $O_m(\eta)$. Under any fixed continuous labeling, the pair is therefore asymptotically controlled. Its quadratic structure is $(t_D, t_P)/2$.*

Proof. Let $\mathcal{R} = D/P$ and let $q : J \to \mathcal{R}$ send (x, c) to the class of x. Choose the based section $s : \mathcal{R} \to D$ and write

$$je = d_D s - s\bar{d}.$$

Then $\tau = (s, -e) : \mathcal{R} \to J$ is a chain section of q. The kernel of q is $\mathrm{Cone}(1_P)$, with contraction

$$k(x, c) = (0, x), \qquad dk + kd = 1$$

after identifying the first coordinate in the kernel with P. Consequently

$$h = k(1 - \tau q), \qquad dh + hd = 1 - \tau q. \tag{14.2}$$

These maps use only a simplex and its faces, so are local.

The composite $F = \Psi q^* : \mathcal{R}^{m-*} \to D$ is the relative cap map of lemma 9.4; let b be its controlled inverse. The shifted dual of (14.2) shows $\Psi \simeq F\tau^*$ and $q^*\tau^* \simeq 1$. Therefore q^*b is an inverse for Ψ. The inverse homotopies use a fixed number of the preceding local maps and homotopies, giving a bound $C_m\eta$. Relabeling uses its continuity modulus, and halving the relative symmetric structure gives the full rational quadratic structure. □

14.2 The tensor for a union

Suppose a closed oriented combinatorial manifold W is cut into A and B along their common boundary P. We now use A, B, P also for their chain complexes. Let j_A, j_B be the boundary inclusions and choose fundamental chains satisfying

$$z_W = z_A + z_B, \qquad dz_A = j_A z_P, \qquad dz_B = -j_B z_P.$$

The algebraic union has underlying complex

$$V = \mathrm{Cone}((j_A, -j_B) : P \to A \oplus B).$$

Let e_A, e_B be the inclusions of the two top complexes. Put $a = e_A j_A$, $b = e_B j_B$, and let $s : P \to V$ be the degree-one inclusion of the suspended interface. Then

$$ds + sd = a - b.$$

With tensor signs understood, put

$$K = s \otimes a + b \otimes s, \qquad K_{\mathrm{sym}} = \tfrac{1}{2}(K + T_V K T_P). \tag{14.3}$$

The homotopy identity implies

$$dK + Kd = a^{\otimes 2} - b^{\otimes 2}.$$

Therefore the symmetric tensor

$$t_V = e_A^{\otimes 2} t_A + e_B^{\otimes 2} t_B - K_{\mathrm{sym}} t_P \tag{14.4}$$

is a cycle. The interface correction is necessary: without it, the two relative boundary terms would remain as different copies of t_P.

Lemma 14.2 (The controlled union has the closed manifold structure)**.** *With common triangulations and diagonals, the Poincaré union defined by (14.4) is controlled-chain-equivalent to the closed manifold's complex with tensor t_W. The same is true of their quadratic structures after taking halves.*

Proof. The collapse $p : V \to C_*(W;\mathbb{Q})$ identifies the two copies of every interface simplex and kills the suspended interface. Its kernel is a copy of $\mathrm{Cone}(1_P)$ and has the local contraction k above. For a degreewise based section r of p, correct its chain-map defect by

$$\ell = r - k(dr - rd).$$

The defect $dr - rd$ lies in the kernel, and the contraction equation gives $d\ell = \ell d$ and $p\ell = 1$. Then $k(1 - \ell p)$ contracts $1 - \ell p$, so p is a local chain homotopy equivalence.

Since $ps = 0$, the correction in (14.4) vanishes after applying $p \otimes p$. The two top fundamental chains add to z_W, and the common diagonal therefore gives

$$(p \otimes p)t_V = t_W \tag{14.5}$$

exactly. The cap inverse of W and the equivalence p show that t_V is Poincaré. All maps and corrections involve cells at distance $O_m(\eta)$; a fixed continuous labeling makes their propagation tend to zero. Equation (14.5) identifies symmetric structures, and the rational norm identification gives the assertion for quadratic structures. This is the rational chain model of the Poincaré union construction in [Ran80I, Section 3]. □

15 Scalar homotopies and evaluation of the boundaries

15.1 A homotopy of labels gives a controlled cobordism

The underlying manifold and its chain complexes need not change when the control map changes. The issue is that the identity matrix between two differently labeled complexes need not have small propagation. A cylinder, with a fine time subdivision, is what implements the change of labels in L-theory.

Lemma 15.1 (Changing scalar control by a fixed homotopy)**.** *Let W be a compact closed oriented combinatorial m-manifold. Let Y_i be its fixed ordered triangulations, with mesh $\eta_i \to 0$, and use their prescribed fundamental tensors. For a continuous map*

$$c : W \times [0,1] \longrightarrow X$$

to a compact metric space, the complexes with endpoint labels c_0 and c_1 determine the same rational quadratic L-class. Their images in any fixed exterior quotient also agree.

Proof. Subdivide the time interval with mesh $\lambda_i \to 0$ and triangulate each product prism. Keep the triangulation Y_i and its vertex order on both ends. One total vertex order on the cylinder can restrict to the prescribed orders at the ends, so its Alexander–Whitney diagonal restricts to the exact endpoint diagonals, not merely to homotopic substitutes.

If z_i is the fundamental cycle of W, the product boundary formula gives

$$d(z_i \times [0,1]) = (-1)^m(j_1 z_i - j_0 z_i).$$

Consequently the oriented cylinder chain

$$Z_i = (-1)^m z_i \times [0,1]$$

has boundary $j_1 z_i - j_0 z_i$. Its symmetrized diagonal and those of the two ends satisfy the exact tensor equation for a Poincaré pair. Label a cylinder simplex by the value of c at its barycenter. The relative duality construction of lemma 14.1 supplies an inverse and inverse homotopies whose propagation is bounded by

$$\omega_c(C_m(\eta_i + \lambda_i)) \longrightarrow 0,$$

where ω_c is a continuity modulus for the fixed map c. The chain degrees lie in $[0, m+1]$, independently of i.

Halving the symmetric tensors gives the required quadratic Poincaré cobordism. Its boundary is the endpoint structure at time 1 minus that at time 0, which is precisely the equality of their L-classes. Applying a quotient functor to this completed cobordism preserves the equality. The relative symmetric construction is the one underlying [Ran80II, Propositions 6.1–6.3]. □

No hypothesis that the homotopy preserves an exterior set is required here. We first construct a controlled cobordism over the whole target X, and only then apply the fixed exterior quotient. This order matters.

In our application these are *scalar* homotopies: we have already forgotten the detector and equivariance. We do not claim that the invariant map f is equivariantly homotopic to a homeomorphism of the source sphere.

15.2 The iterated boundary map

Return to $N = n+4$, $X = T \times S^n$, and the radii fixed in section 8. For the first cut put

$$\begin{aligned} A &= T \times (S^n \setminus B(y,r)), \\ B &= T \times \overline{B(y,r)}, \\ Y &= A \cap B = T \times S_r(y). \end{aligned} \tag{15.1}$$

Here $S_r(y) = \{z : d(y,z) = r\}$ is a sphere of dimension $n-1$. The exterior is

$$Z = T \times \{z : d(y,z) \geq R\}.$$

Since $R > r$, we have $Z \subset A$ and $d(Z,B) \geq R - r > 0$. The exterior boundary of proposition 13.1 is therefore defined and gives

$$L_{n+4}(\mathcal{B}_{\mathbf{G},Z}(T \times S^n)) \longrightarrow L_{n+3}(\mathcal{A}_{\mathbf{G},Y}(T \times S^n)).$$

The support-relabeling equivalence identifies the target with the controlled category over $T \times S_r(y)$.

Next identify $S_r(y)$ with a standard S^{n-1} and cover it by two closed hemispheres. Their intersection is the equator S^{n-2}. The ordinary boundary for this cover lowers the L-degree by one and restricts the support to $T \times S^{n-2}$. Repeat this procedure. The first cut and the subsequent $n-1$ cuts together give n boundaries:

$$(n+4, S^n) \longmapsto (n+3, S^{n-1}) \longmapsto \cdots \longmapsto (4, S^0).$$

The notation records the L-degree and the sphere factor remaining in the control space; it is not a map of manifolds.

The two components of $T \times S^0$ are compact and positively separated. An asymptotically controlled matrix eventually has no entries between them. The category therefore splits into the two component categories, and projection to either component is a duality-preserving functor. Choose one component once and for all. The result is

$$\Delta_T : L_{n+4}(\mathcal{B}_{\mathbf{G},Z}(T \times S^n)) \longrightarrow L_4(\mathcal{A}_{\mathbf{G}}(T)). \tag{15.2}$$

The cuts, identifications, orientations, and chosen component are all fixed independently of the sequence index.

With no detector and trivial groups, the same construction gives

$$\Delta_{\mathbf{1}} : L_{n+4}(\mathcal{B}_{\mathbf{1},Z_0}(S^n)) \longrightarrow L_4(\mathcal{V}_{\mathbf{1}}). \tag{15.3}$$

The target has this form because control at a point imposes no propagation restriction: $\mathcal{A}_{\mathbf{1}}(\{*\}) = \mathcal{V}_{\mathbf{1}}$.

Every operation used in the construction is natural for forgetting equivariance and the detector. Writing U for these forgetting functors in the relevant degrees, we obtain

$$U_* \Delta_T = \Delta_{\mathbf{1}} U_*. \tag{15.4}$$

The same argument shows that the boundaries commute with tensoring by the positive permutation forms τ_i. This uses naturality of localization, not a calculation with representatives of individual signature forms.

15.3 Replacing the averaged control by a homeomorphism

The fixed scalar homotopy in (8.8) joins f to the degree-one pinch map s_ρ. Applying lemma 15.1 to $W = S^n \times \mathbb{CP}^2$, with control ignoring the second factor, gives

$$\mathfrak{m}_f = \mathfrak{m}_{s_\rho}. \tag{15.5}$$

Choose an orientation-preserving homeomorphism $g : S^n \to S^n$ and take the source disk and its successive equatorial cuts to be the preimages under g of the target ball and the fixed cutting flag. One can choose this finite flag to be PL with respect to a smooth triangulation of the source sphere. These are the cuts for which the triangulations Y_i were chosen in section 9; this choice precedes all refinements.

Both s_ρ and g have degree one. The degree classification of sphere maps therefore gives a fixed homotopy $s_\rho \simeq g$ for $n \geq 1$ [Hat02, Corollary 4.25]. A second application of lemma 15.1 yields

$$\mathfrak{m}_f = \mathfrak{m}_g. \tag{15.6}$$

All target cuts remain unchanged during these homotopies. The point of replacing f is that the inverse image under g of each target cut is now a predetermined manifold cut. No regularity or manifold structure is required of the level sets of F_0 or f.

15.4 Why the surviving signature is one

Proposition 15.2 (Scalar boundary evaluation)**.** *There is a sign $\varepsilon \in \{1, -1\}$, determined only by the fixed orientation and component conventions, such that*

$$\Delta_{\mathbf{1}}\mathfrak{m}_f = \varepsilon[C_*(\mathbb{CP}^2;\mathbb{Q}), \phi_{\mathbb{CP}^2}/2], \tag{15.7}$$

where the right side denotes a constant point-controlled class. In particular,

$$\sigma_{\mathrm{tail}}(\Delta_{\mathbf{1}}\mathfrak{m}_f) = [(\varepsilon, \varepsilon, \ldots)]. \tag{15.8}$$

Proof. By (15.6), it suffices to compute on the model controlled by g. The first cut decomposes the source sphere into a disk D and its complement. After taking the product with $\mathbb{CP}^2$, the two pieces meet along

$$\partial D \times \mathbb{CP}^2.$$

The fixed triangulations respect these pieces and the subsequent flag. Their common diagonal and compatible fundamental chains give the controlled Poincaré pairs of lemma 14.1. By lemma 14.2, their union is the prescribed closed-manifold class, including its full rational quadratic structure.

The first interface is positively separated from the exterior. Thus lemma 13.2 identifies the first localization boundary with the interface class, oriented as the boundary of the piece called A in (15.1). Each subsequent boundary replaces the current sphere interface by its next equator and retains the $\mathbb{CP}^2$ factor. After n boundaries the interface is

$$S^0 \times \mathbb{CP}^2.$$

Projection onto the chosen component leaves exactly one copy of $\mathbb{CP}^2$. Boundary-orientation conventions contribute a sign ε, but every choice was fixed before the sequence varied, so this sign is independent of i.

The resulting triangulations of this copy of $\mathbb{CP}^2$ may become finer with i. At point control, subdivision and its chain inverse have no spatial restriction. They identify the resulting sequence with a constant Poincaré model. Its middle-dimensional intersection form is $\langle 1\rangle$ by (3.1). The quadratic structure is half the symmetric structure, so its symmetrization is still the form $\langle 1\rangle$, not $\langle 1/2\rangle$. Its ordinary signature is 1. Reversing the orientation changes the signature to -1, proving both formulas. □

For $n = 1$, there is only the first cut. Its interface already consists of two copies of $\mathbb{CP}^2$, with the endpoint orientations induced by a one-dimensional disk. Keeping both components and adding their signatures could cancel them. The component projection in (15.2) is what keeps one copy. The same issue is present at the last stage in every dimension.

16 The conclusion and the Hilbert–Smith reduction

16.1 The signature-one class

The realization and scalar comparison of sections 12 and 15 give

$$\mathfrak{d} \in L_{n+4}(\mathcal{B}_{\mathbf{G},Z}(T \times S^n)), \qquad U_*\mathfrak{d} = \mathfrak{m}_f.$$

The algebraic obstruction applies not directly to $\mathfrak{d}$, which still has an exterior, but to its iterated boundary. Define

$$\alpha = \varepsilon \Delta_T \mathfrak{d} \in L_4(\mathcal{A}_{\mathbf{G}}(T)), \tag{16.1}$$

where ε is the sign from proposition 15.2. Naturality and scalar evaluation give

$$\begin{aligned} U_*\alpha &= \varepsilon \Delta_{\mathbf{1}} U_* \mathfrak{d} \\ &= \varepsilon \Delta_{\mathbf{1}} \mathfrak{m}_f \\ &= [C_*(\mathbb{CP}^2; \mathbb{Q}), \phi_{\mathbb{CP}^2}/2]. \end{aligned}$$

The unnormalized coefficient-of-identity trace identifies the signature of an equivariant rational form with the ordinary signature of its underlying scalar form. Tail extraction commutes with this forgetting operation. Therefore

$$\sigma_{\mathrm{tail}}(\alpha) = [(1, 1, 1, \ldots)]. \tag{16.2}$$

On the other hand, T is a fixed compact free C_p-space, and the groups G_i are cyclic p-groups with index-p subgroups P_i. The divisibility theorem, theorem 7.3, applies to exactly this category and gives

$$\sigma_{\mathrm{tail}}(\alpha) \in p(\mathbb{Z}^{\mathbb{N}}/\mathbb{Z}^{(\mathbb{N})}).$$

An integer sequence representing the constant tail 1 cannot have every sufficiently late entry divisible by p. This is the required contradiction for a faithful action on a boundaryless manifold of positive dimension.

16.2 Dimension zero and manifolds with boundary

A connected zero-dimensional manifold is a point, and any action on it has the whole group as kernel. For a connected manifold M with boundary and positive dimension, its interior M° is dense and connected. Density follows from the half-space coordinate charts at boundary points. For connectedness, each sufficiently small boundary half-ball has connected interior and therefore meets only one interior component. Adjoining each boundary point to that component would turn a separation of the interior into a separation of M. Since M is connected, there can be only one component.

A homeomorphism preserves the interior. One intrinsic way to see this is through local homology with coefficients $\mathbb{Z}/2$: at an interior point, the top local homology group is $\mathbb{Z}/2$, whereas at a boundary point it is zero. Homeomorphisms preserve these groups. Restriction of an action to M° is therefore defined. If an element fixes M° pointwise, continuity makes it fix its closure M. Hence a faithful action remains faithful on the interior. The exclusion in the boundaryless case therefore extends to manifolds with boundary.

Proof of theorem 1.1. For positive-dimensional manifolds without boundary, a faithful action produces the class α in (16.1), whose signature tail is (16.2). This contradicts the divisibility theorem, theorem 7.3. A connected zero-dimensional manifold is a point. The preceding restriction-to-the-interior argument handles manifolds with boundary. □

16.3 What is needed to deduce the Hilbert–Smith assertion

A Hausdorff topological group is *locally compact* if every point has a neighborhood with compact closure; by translation it is enough to check this at the identity. A *Lie group* is a finite-dimensional smooth manifold equipped with smooth multiplication and inversion. Saying that a topological group is a Lie group *with its given topology* means that the manifold topology is its original group topology, not merely that its underlying abstract group happens to be isomorphic to some Lie group.

A group has *no small subgroups*, abbreviated NSS, if some neighborhood of its identity contains no nontrivial subgroup. In contrast, $\mathbb{Z}_p$ has arbitrarily small nontrivial subgroups H_j.

The passage from p-adic exclusion to Hilbert–Smith uses additional topological-group results. The particular form needed is the following.

Foundational theorem 16.1 (Compact reduction, Newman, and no small subgroups)**.** *The standard locally compact Hilbert–Smith reduction has these consequences.*

(a) Compact reduction. *If faithful continuous $\mathbb{Z}_p$-actions on a fixed connected manifold are excluded for every prime p, then every compact group acting continuously and faithfully on that manifold is a Lie group.*

(b) Uniform Newman property. *Fix a compatible metric on a connected manifold and a nonempty open subset V. There exists $\epsilon > 0$ such that a compact Lie group action satisfying*

$$d(gx, x) < \epsilon \quad \text{for every } g \text{ and every } x \in V$$

is trivial. The same ϵ works for all such compact Lie groups and actions.

(c) Gleason–Yamabe consequence. *A locally compact Hausdorff group with no small subgroups is a Lie group in its given topology, under the second-countability convention used here.*

The compact p-adic reduction and the uniform Newman formulation are described in [Pa19]; the no-small-subgroups conclusion is part of the structure theory developed in [Gle52, Yam53]. These are external foundational inputs. In particular, part (a) must not be replaced by the false assertion that every non-Lie compact group contains a copy of $\mathbb{Z}_p$. For example, the infinite product $\prod_{j\geq 1} C_p$ has exponent p, whereas the additive group $\mathbb{Z}_p$ is torsion-free. The reduction uses the restrictions imposed by a manifold action as well as group structure.

Proof of theorem 1.2. Let L be a second-countable locally compact Hausdorff group acting continuously and faithfully on a connected manifold. By theorem 1.1 the p-adic exclusion holds for every prime. Every compact subgroup of L acts faithfully and is therefore a Lie group by part (a).

Choose a nonempty relatively compact coordinate neighborhood V and let $K = \overline{V}$. Let ϵ be the Newman constant. Joint continuity of the action, uniformly over the compact set K, supplies an identity neighborhood $O \subset L$ such that

$$d(gx, x) < \epsilon \qquad (g \in O,\ x \in K).$$

Choose an identity neighborhood W with compact closure contained in O. If a subgroup S is contained in W, its closure $\overline{S}$ is a compact subgroup contained in O. It is a compact Lie group, and its action satisfies the small-displacement condition on V. By part (b), its action is trivial. Faithfulness gives $\overline{S} = \{1\}$ and therefore $S = \{1\}$.

Thus L has no small subgroups. Part (c) makes L a Lie group in its original topology. □

17 Uniformity requirements and an index of the construction

17.1 The quantifiers that the construction requires

There are several different uses of "small" in this argument. They should not be conflated.

A subgroup that is small once. The subgroup H is chosen so that its elements move chart points by at most a fixed small constant D. This is enough to define the coordinate average, preserve the necessary chart buffers, and build the fixed detector region. The number D does not subsequently tend to zero.

A kernel sequence that becomes small. The subgroups $K_i \subset H'$ shrink to the identity. Their displacements on the fixed compact region tend uniformly to zero. This makes products of chosen quotient representatives close to the representative of the product where control is needed.

A mesh and propagation that become small. The meshes η_i, approximation errors μ_i, and uniform primitive bounds ϵ_i tend to zero. A morphism in the asymptotic category needs this spatial smallness. Small numerical coefficients alone do not provide it.

Data that must stay fixed. The compact detector T, its C_p-action, the finite trivializing cover used in the nilpotence proof, the radii r, R, t, u, the sphere cuts, and the number of chain degrees do not depend on i. In particular, the nilpotence exponent is a fixed integer, not a number allowed to grow with $|G_i|$.

Data allowed to grow. The numbers of simplices, group orders, ranks of modules, numbers of summands in a group average, and sizes or denominators of rational coefficients may grow without restriction. Such growth is harmless only because every term in each relevant sum has a common propagation bound tending to zero.

Only boundedly many compositions at a time. The exterior-band argument concerns words of a fixed bounded length in the primitive maps. A fixed number of $o(1)$ propagation errors is still $o(1)$. Nothing in that argument permits a number of compositions increasing arbitrarily with i.

Exact algebraic equations in the quotient. An equation in the exterior category is not justified by saying that its error is numerically small. The defect must be an exact matrix factorization through an exterior-supported object, with both factors asymptotically controlled. The compression formulas exhibit the required factors through Q or Q^*.

One splitting of a sequence object. The homotopy idempotent is split in the homotopy category of the sequence category itself. An arbitrary list of finite-dimensional splittings, one for each i, would not automatically have the uniform control or bounded degree range required here.

An integer conclusion only after removing the exterior. The class $\mathfrak{d}$ is a germ near a controlled region and initially lives in an exterior quotient. The iterated boundary removes the exterior and puts α in the ordinary controlled category over T. Only then is the coordinate signature tail taken. Rational coefficients in the complexes do not alter the integer-valued signature target.

17.2 Objects, maps, and their roles

Notation	Meaning and role
M	The manifold on which the hypothetical faithful $\mathbb{Z}_p$-action is given. It is not assumed triangulable.
$H_j = p^j\mathbb{Z}_p$	The nonzero closed subgroups of $\mathbb{Z}_p$; they are also open and form an identity-neighborhood basis.
H, H'	A small fixed subgroup $H = H_j$ and its index-p subgroup $H' = H_{j+1}$.
K_i	A sequence of subgroups of H' shrinking to the identity.
G_i, P_i, q_i	The finite groups $G_i = H/K_i$, their index-p subgroups $P_i = H'/K_i$, and their orders $q_i = \lvert G_i\rvert$.
F_0	The Haar average of coordinate vectors. It is H-invariant and close to the coordinate identity, but need not be injective.
s_ρ, f	A degree-one pinch map and the invariant sphere control map $f = s_\rho F_0$, extended by infinity.
Z^+	A fixed compact invariant region containing the relevant support of f and containing no H-fixed points.
T, $\widetilde{\rho}$	A fixed odd-dimensional unit sphere with free scalar C_p-action, and the equivariant detector from Z^+ to T.
$r < R < t < u$	Fixed buffer radii: the first cut, the exterior threshold, the compression threshold, and an outer detector buffer.
Z_0, Z	The discarded exterior in sphere control and in detector–sphere control, respectively.
$W = S^n \times \mathbb{CP}^2$	A smooth auxiliary manifold built from a compactified coordinate sphere. Its $\mathbb{CP}^2$ factor supplies signature one.
$N = n + 4$	The Poincaré dimension and the common upper chain degree.
Y_i, D_i	Shrinking-mesh triangulations of W and their finite rational simplicial chain complexes.
z_i, ϕ_i, b_i	Fundamental cycles, strictly symmetric cap dualities, and controlled homotopy inverses.
$a_{g,i}$	Chain approximations to completed representatives of $g \in G_i$. They need not constitute a strict group action.
$B_{g,h}$	Uniform graph-controlled homotopies witnessing multiplication of quotient representatives on chains.
U_g, V_g	The local isometry homotopies and the resulting adjoint-compatibility homotopies.

Notation	Meaning and role
F_i, C_i	The far-supported subcomplex of D_i and the compressed quotient complex $C_i = D_i/F_i$.
π, σ, Q	Quotient projection, degreewise based section, and far projection $Q = 1 - \sigma\pi$. The section is not generally a chain map.
A_g	The compressed representative $\pi a_g \sigma$, a chain map only modulo the specified exterior defects.
$\mathcal{C}_i$	The free-sheet complex $C_i \otimes \mathbb{Q}[G_i]$, carrying a strict free left G_i-action on its basis.
E_i	The averaged graph homotopy idempotent $q_i^{-1} \sum_g A_g \otimes R_{g^{-1}}$.
D, ϕ_D	The split corner and its normalized duality $q^{-1} r \Phi r^*$. This D is not the original triangulated complex D_i.
$\mathfrak{d}$	The corner's controlled quadratic L-class before the exterior is removed.
U	Forgetting equivariance and detector labels while retaining sphere control; it is unrelated to the homotopies U_g.
$\mathfrak{m}_c$	The scalar exterior manifold class of the fixed complexes labeled by a control map c.
τ_i, ν	The positive quotient-permutation form and the operation $\nu = p\,\mathrm{id} - \tau \otimes (-)$, nilpotent over fixed free T.
Δ_T	The n successive localization boundaries followed by a choice of one final component.
α	The ordinary controlled degree-four class $\varepsilon \Delta_T \mathfrak{d}$ to which signature divisibility applies.

17.3 Terminology by location

Terms	Defined or developed in
Faithful/effective action, free action, free point, orbit, stabilizer, kernel, fixed set, invariant and equivariant map	section 2
p-adic integer, valuation, ultrametric, clopen subgroup, profinite group, Haar measure and averaging	section 2
Orientation, degree, one-point compactification, $\mathbb{CP}^2$, triangulation, combinatorial manifold, PL, mesh, star, link	section 3
Chain complex, augmentation, acyclic complex, chain homotopy, mapping cone, relative chain complex, fundamental chain	section 3
Group algebra, free and projective module, representation, character, signature, Lagrangian, metabolic form, Witt group	sections 3 and 6
Category, additive category, functor, full faithfulness, duality, propagation, support, ideal, exterior quotient	section 4
Karoubi filtration, idempotent completion, homotopy category, homotopy idempotent	sections 4, 5 and 12
Symmetric and quadratic Poincaré complex, Poincaré pair, cobordism, L-group, decoration, spectrum, negative K-group	section 5
Tail, eventual equality, equivariant signature, cyclotomic polynomial, permutation form	sections 6 and 7
Covering, sheet, local trivialization, partition of unity, detector	sections 7 and 8
Carrier, dual block, cap duality, graph type, compression, controlled exterior witness	sections 9 to 11
Localization boundary, Mayer–Vietoris, Poincaré union, scalar normalization	sections 7 and 12 to 14
Locally compact group, Lie group, no small subgroups, Newman property	section 16

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